

Biology Notes
Class X
ICSE





INDEX

NAME: Gauri Singh STD.: X SEC.: D ROLL NO.: 42 SUB.: Biology

S. No.	Date	Title	Page No.	Teacher's Sign / Remarks
1.	13.4.22	Structure of Chromosomes	01	V Good neat work 9/4/22
2.	15.4.22	Cell Cycle and Cell Division	04	
3.	26.6.22	Absorption by Roots	16	
4.	2.8.22	Genetics	30	V Good clear work 14/10/22
5.	15.8.22	Photosynthesis	37	
6.	28.9.22	Transpiration	48	
7.	29.9.22	Circulatory System — Complete left over part	58	
8.	21.12.22	The Excretory System	80	
9.	24.12.22	The Endocrine System	87	
10.	28.12.22	The Reproductive System	94	
11.	03.1.23	The Nervous System	107	
		— (Question/Answers)	107	
		— (Notes)	111	
12.	17.1.23	Pollution	115	
13.	30.1.23	MCQs	117	
14.		Population	129	

Cell only for KCV

classmate
Date
Page 01

13.4.22

Structure of chromosomes

- Chromosomes play a vital role in hereditary variation and evolutionary development of the species.
- They were discovered by a German scientist Walther Flemming in 1882.
- Chromosomes are highly coiled and condensed chromatin fibres seen only in the nucleus only during cell division.
- Each chromosome has two sister chromatids attached by a small constriction called centromere.
- A chromosome consists of about 40% DNA and 60% histone proteins
- The DNA strand winds around a core of eight histone molecules. This complex is known as a nucleosome.
- Based on function, two kinds of chromosomes are found in the human body
  i) Autosomes (Non-sex chromosomes)
  ii) Heterosomes or Allosomes (Sex chromosomes)

- Chromatin fibres are thread-like structures formed of DNA and Histones.

Structure of DNA

- DNA is a macromolecule of nucleotides composed of two complementary strands twisted around each other.

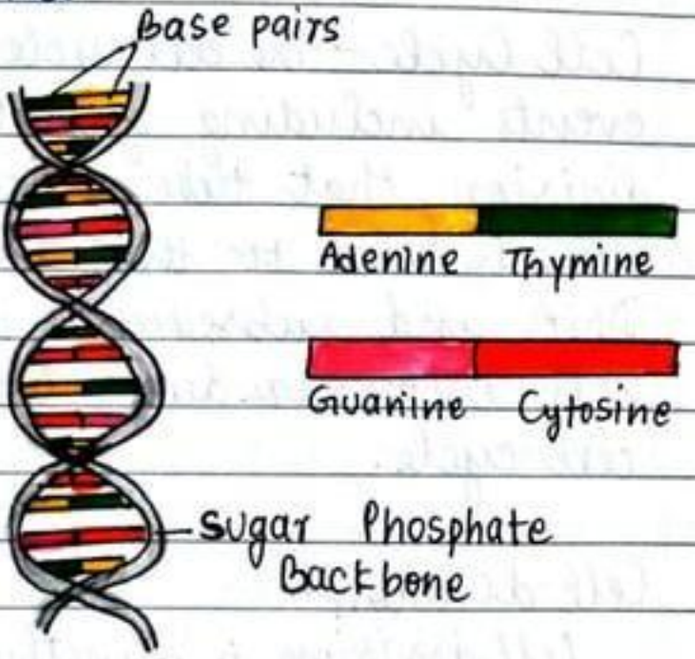
- A nucleotide is made of phosphate, deoxyribose sugar and a nitrogenous base.
  ↓
  There are foure nitrogenous bases – Adenine(A), Cytosine(C), Thymine(T) and Guanine(G)

- Pairing of the complementary base pairs occurs between A and T with two hydrogen bonds and between G and C with three hydrogen bonds.

Genes – Genes are specific regions on chromosomes which are composed of DNA. They determine hereditary characters and are responsible for transmission of characters from parents to offsprings.

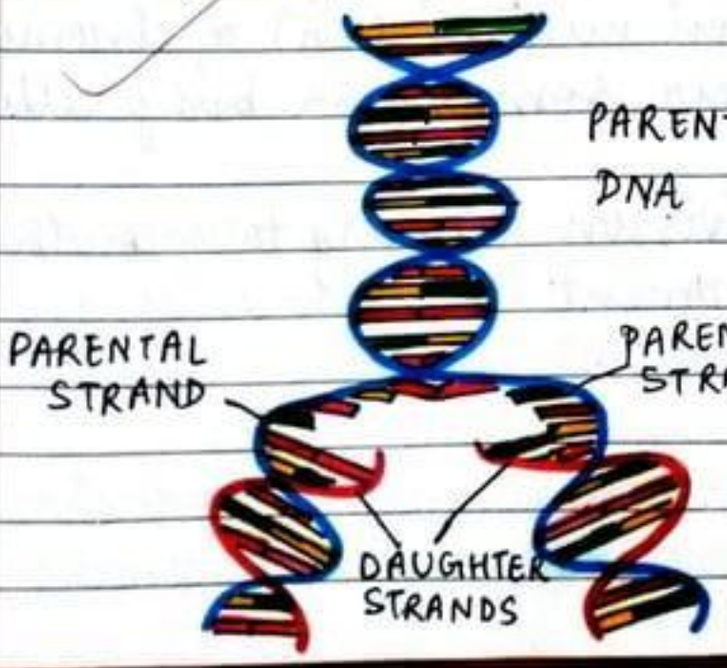
Scientists have identified

around 1,30,000 genes in human chromosomes.



Formation of New DNA - Formation of new DNA occurs by the process of replication. It is a complex process which helps in the conservation of one old strand (parental strand) and synthesis of new strand.

Thus replication of DNA is a semi-conservative model.



16.04.22

Cell Cycle and Cell Division

Cell Cycle - The cell cycle is a series of events including growth ~~division~~ and division, that take place in a cell which lead to the duplication of its DNA and subsequent division of the cell into ~~two~~ daughter cells is called cell cycle.

Cell division -
Cell division is a method by which new cells are originated from pre-existing cells.

These new cells need to be produced for growth, development, replacement repair and reproduction.

~~All~~ All organisms of ~~a species~~ contain an even number (2n) of chromosom-es in their somatic or body cells.

Cell division leading to growth, development, repair and replacement is called MITOSIS.

Cell division leading to production of gametes (sex cells) is called MEIOSIS

Phases of Cell Cycle —

Interphase (Resting or non-dividing phase)
Mitotic phase (Dividing phase)

INTERPHASE —

It is the non-dividing preparatory phase during which a cell grows to its maximum size (by increase in volume of nucleus and nucleolus). The synthesis of DNA takes place due to which ~~so~~ the chromatin material also duplicates.

Interphase is divided into three sub-stages —

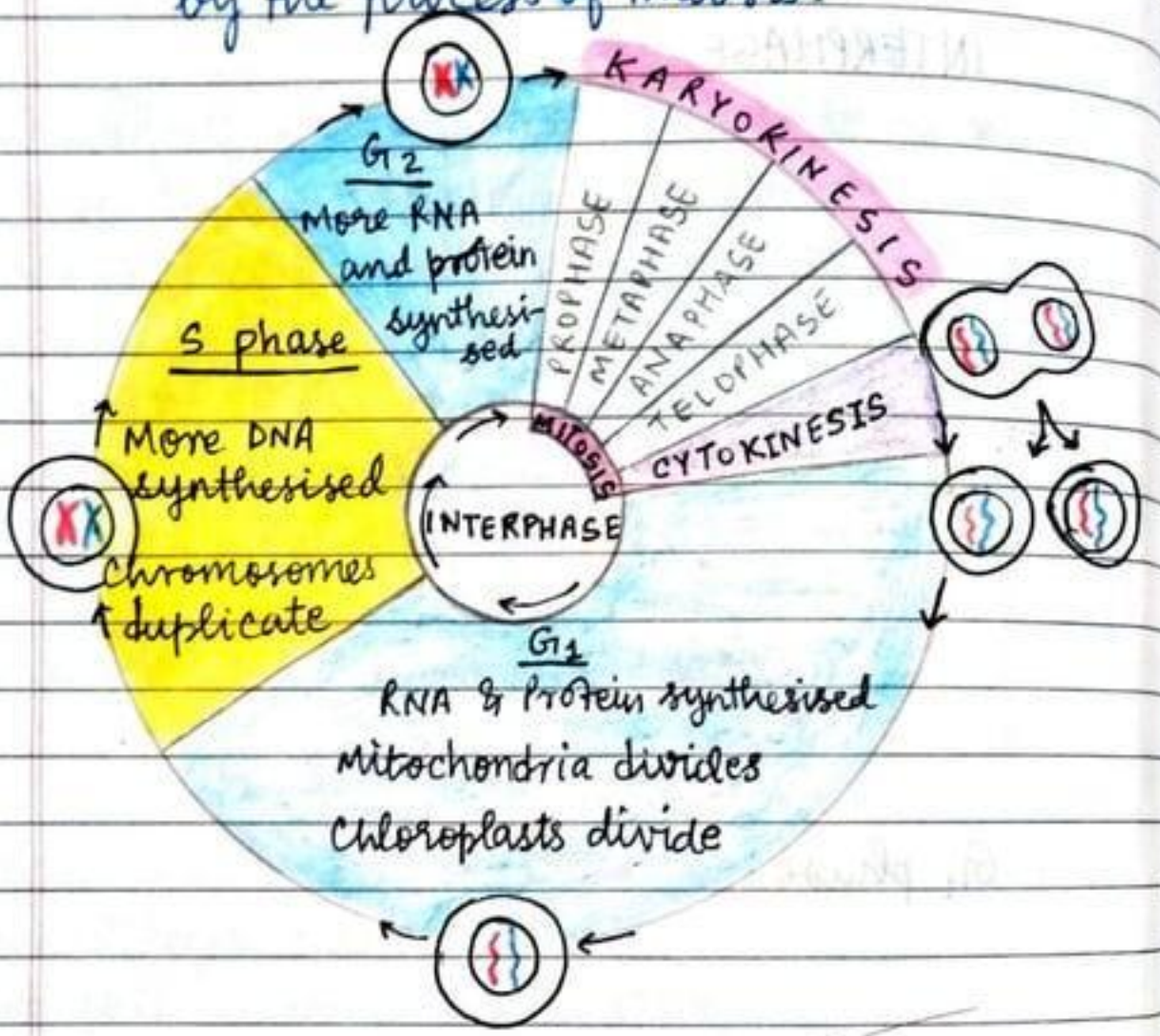
G_1 phase: Volume of cytoplasm increases, RNA and protein synthesis take place. The cells which stop dividing enter in G_0 phase.

S phase: Synthesis of DNA takes place

G_2 phase: Necessary spindle proteins are synthesized, synthesis of RNA and enzymes takes place.

The cell is quite active metabolically during this phase and several

processes occur at this time (like DNA synthesis). Thus no alternative is left for the cell except to divide by the process of mitosis.



MITOTIS

It is the dividing phase during which nucleus as well as cytoplasm divide into two daughter cells. It is studied under two parts —

1. Karyokinesis
2. Cytokinesis

Karyokinesis - (Division of Nucleus)

It has four stages:

(a) **PROPHASE** It begins with the appearance of chromosomes which have become short, thick and clearly visible inside the nucleus.



- Each chromosome gets duplicated by splitting longitudinally to form two chromatids which are attached to each other by centromere.

- The centrosome splits into two and then duplicates the centrioles which move towards opposite poles of the cell.

- Each centriole is surrounded by asters (centrioles with astral rays). A number of spindle fibres appear between two daughter centrioles.

- In plant cell, centrosome is absent. So the formation of spindle fibres take place from microtubules of the cell.

- The nuclear membrane and nucleolus disappear
- The chromosomes start moving towards equator of the cell

(b) METAPHASE

- The chromosomes get ~~at~~ arranged at the equatorial plate of the cell.
- Each chromosome gets attached to a spindle fibre by its centromere

(c) ANAPHASE

Centromere divides and two sister chromatids of each chromosome separate and are drawn apart towards opposite poles to form two groups of chromosomes.

(d) TELOPHASE

- Each chromatid or daughter chromosome turns into a network of chromatin threads.
- Nuclear membrane and nucleolus reappear.

Animal Cell

Plant Cell

classmate
Date
Page 09

Centrioles duplicated and moving apart

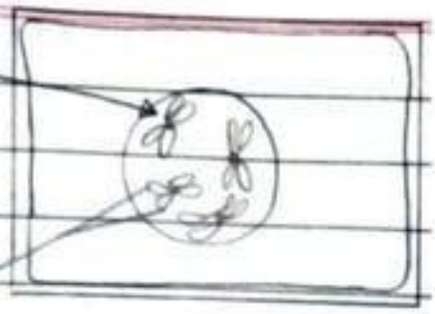
Aster



Centromere

EARLY PROPHASE

Chromatids (Duplicated Chromosome)

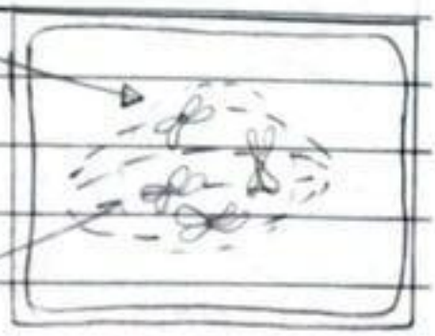


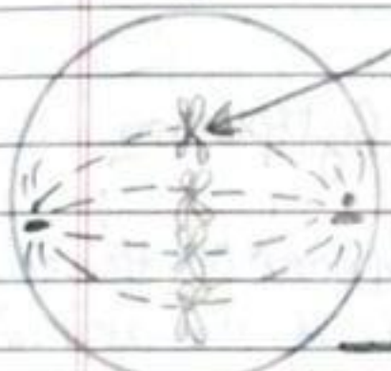


Nuclear membrane disappears

LATE PROPHASE

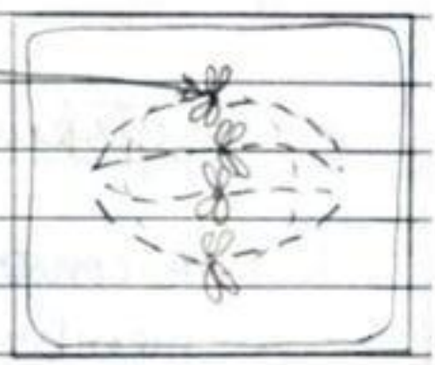
spindle fibres

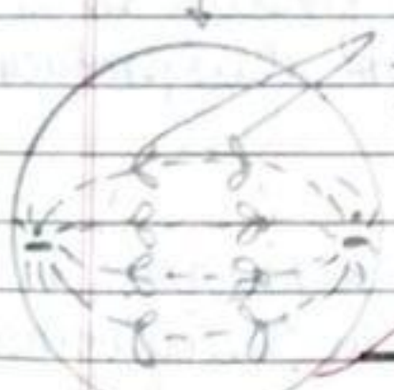




Chromosomes line along the equatorial plane

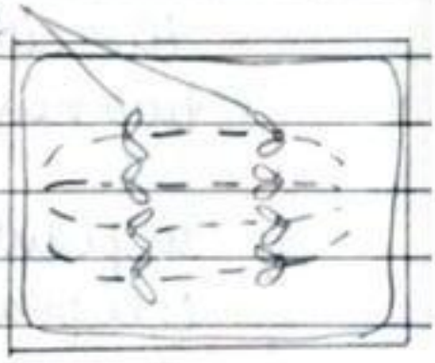
METAPHASE

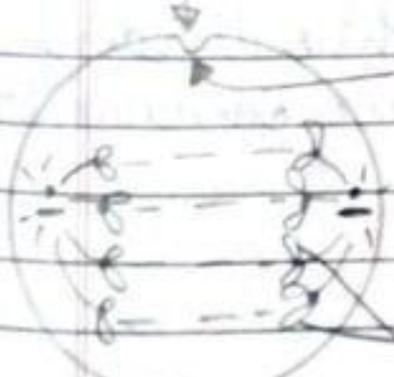




Sister chromatids separate and move to opposite poles

EARLY ANAPHASE

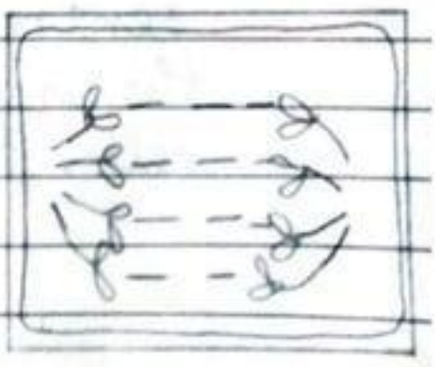




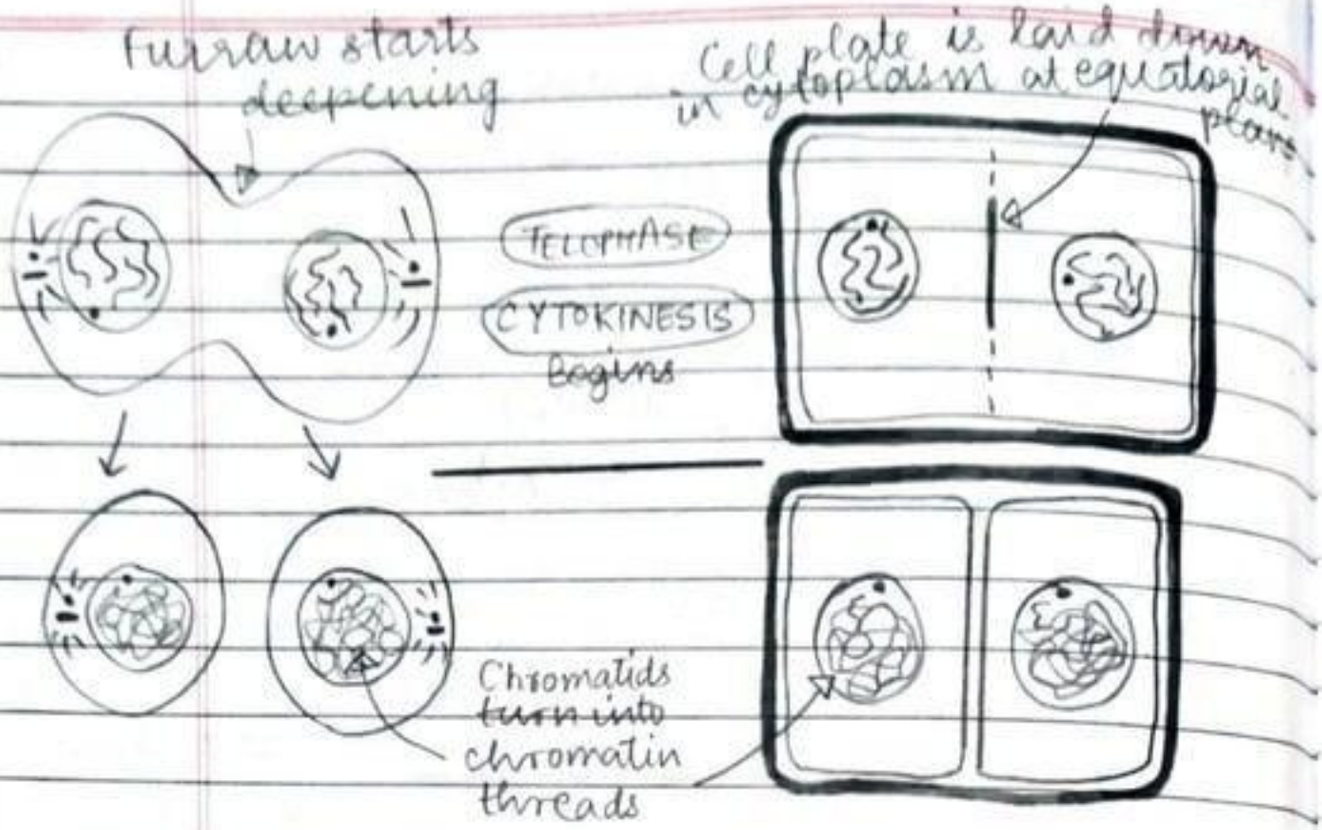
Furrow in cytoplasm

LATE ANAPHASE

Daughter chromatids



classmate
Date
Page



Cytokinesis - Division of Cytoplasm

A constriction (furrow) appears in the cell membrane at the middle of the animal cell, which deepens and finally splits the cytoplasm into two, thus producing two new cells.

In plant cell, instead of a furrow, a cell plate is laid down in the cytoplasm at equatorial plate. It grows from centre to periphery, thus dividing the original cell into two ~~the~~ daughter cells.

NOTE

In mitosis, two identical daughter cells are produced by division of one parent cell. The main aspect of this division is that the same normal chromosome number is maintained at each division of the cell. Hence, it is also called Equational Cell Division.

MITOSIS IN ANIMAL CELL	MITOSIS IN PLANT CELL
Centriole is present and it divides into two in early prophase.	centriole is absent
Aster is formed from each centriole	Aster is not formed
Cytokinesis occurs by furrow formation	Cytokinesis occurs by cell-plate formation
Cell division occurs in tissues throughout the body.	Cell division occurs mainly in growing regions.

Significance of Mitosis—

i) Helps in growth of animals and plants
ii) Replaces old and dead cells
iii) Repairs damaged tissues

iv) Helps in asexual reproduction in unicellular organisms.
v) Maintains same number of chromosomes in daughter cells.

MEIOSIS

It is the kind of cell division that produces gametes or sex cells.

It takes place in the reproductive organs i.e., testes to produce sperms and ovary to produce egg (ovum in humans).

In plants, it occurs in anthers to produce pollen grains and in ovules to produce egg.

The number of chromosomes in this type of cell division is reduced to half in the daughter cells.

As the cells in an organism are diploid (2n), i.e., they contain ~~2n~~ pairs of chromosome, the formation of gametes takes place by reductional division. The gametes are haploid (n) so that when two haploid gametes fuse (egg and sperm), the resulting zygote has

full diploid set of chromosomes.
In this way, the chromosome number is maintained constant from one generation to the next generation by meiosis.

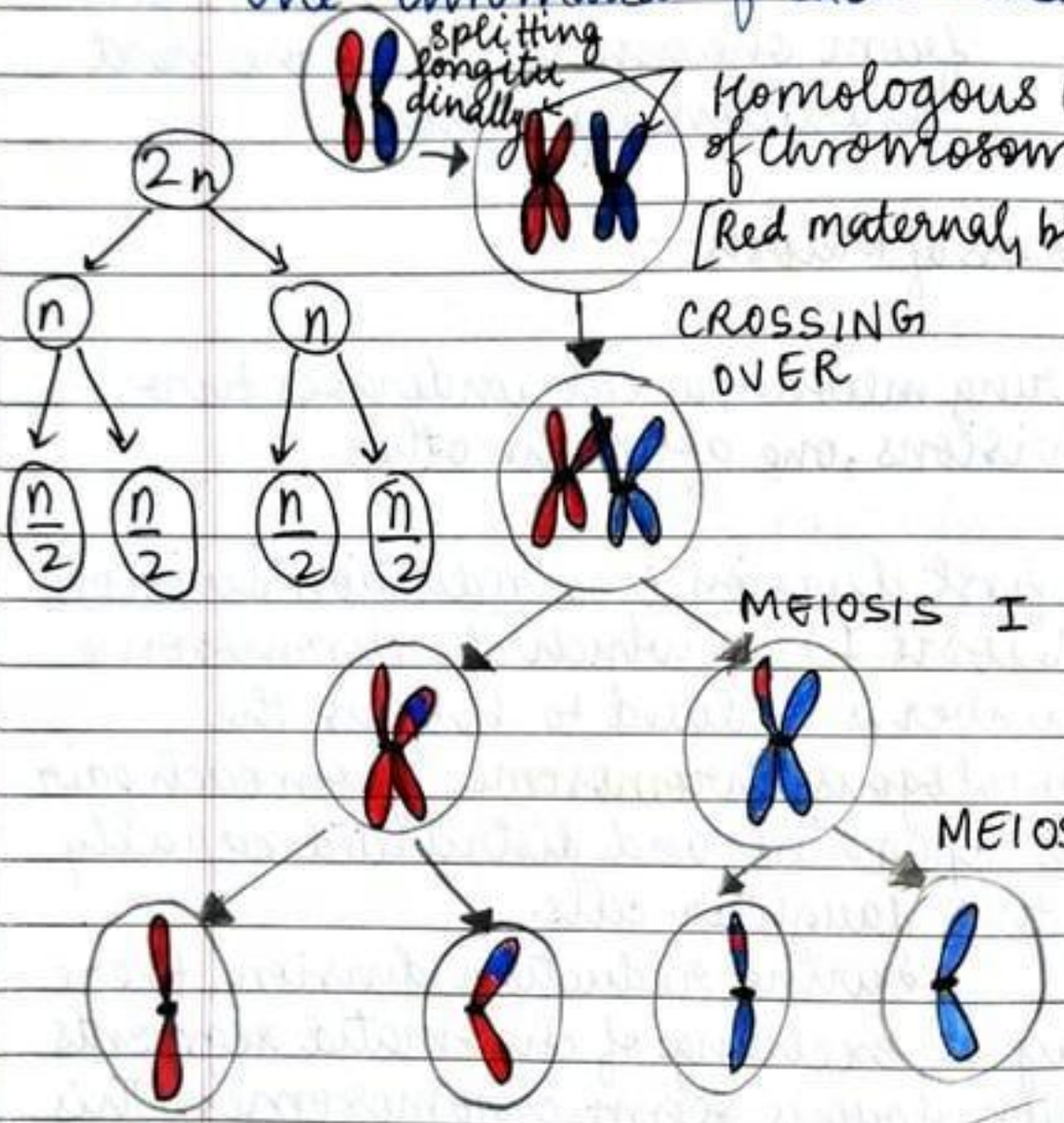
Process of Meiosis —

During meiosis, a cell undergoes two divisions, one after the other.

1. The first division is a reduction division (Meiosis 1) in which the chromosome number is reduced to half, as the homologous chromosome from each pair gets separated and distributed equally in two daughter cells.
During reduction division, there may be exchange of chromatid segments of homologous ~~segm~~ chromosomes. This is called crossing over. As a result of crossing over, exchange of genes occurs and produces a new gene combination in the offspring.

2. The second division is simple mitotic division (Meiosis II) in which each chromosome of daughter cells (formed after Meiosis I) breaks up into two chromatids. The two

daughter cells further divided to form ~~for~~ four daughter cells containing one chromatid of each kind.



Significance of Meiosis—

1) In sex cells (gametes), the chromosome number is halved so that on fertilization, the normal number (2n) is restored.

2) ~~Meiso~~ Meiosis helps in mixing up of genes which increases the genetic vigour.

iii) Helps in causing variations which further leads to evolution.

MITOSIS	MEIOSIS
It occurs in the somatic/vegetative cells of the body.	It occurs in reproductive cells.
It provides growth, repair and replacement.	It occurs for gamete formation
It results in the formation of two identical daughter cells.	It results in the formation of four ~~duplex~~ dissimilar daughter cells.
The number of chromosomes in the daughter cells remains the ~~same~~ same as parent cell. (EQUATIONAL DIVISION)	The number of chromosomes in the daughter cells is reduced to half of the parent cell. (REDUCTIONAL DIVISION)
It involves ~~two~~ single nuclear division	Meiosis involves two successive nuclear divisions - Meiosis I and Meiosis II.

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26.06.22

Ch- Absorption by Roots

Plants need water for a number of physiological activities. and Water constitute about 90% weight of a plant.

Roots absorb water and mineral nutrients from soil by root hair.

Need of water and minerals for plants -

1. Photosynthesis - Water is used by the green parts of the plant as a raw material for the synthesis of glucose.

2. Transpiration - A large quantity of water is evaporated as water vapour during transpiration, for cooling in hot weather, for producing a suction force, etc.

3. Transportation - Water is a universal solvent which causes transportation of substances in solution from roots upward to the shoot (mineral salts) or from leaves to other parts (sugar).

4. Mechanical stiffness - Water provides turgidity which is necessary for the

stiffness of plant tissues.

5. Minerals- Act as building blocks for many compounds synthesised by plants. Some mineral & elements are required by plants as constituents of cells and cell organelles, and synthesis of a variety of compounds. These minerals are absorbed by roots ~~ont~~ only.

Characteristics of roots for absorbing water -

1. Roots have very large surface area - they have many rootlets and each rootlet has a number of root hairs at its tip. Their large number increases the surface area of the root for increased absorption of water and minerals.

2. Roots contain concentrated cell sap - The cell sap of root hairs is more concentrated as compared to the surrounding soil water. This facilitates the water to enter into the root hairs by the process of osmosis, due to increased osmotic pressure of the cell sap.

3. **Root hairs have thin cell wall**

Thin cellulose cell walls of the root hairs increases its permeability to let soil water enter more freely into the root hair cell.

The cell membrane of root hair is also very thin and semi-permeable which allows water molecules to pass through, but not the larger molecules of dissolved salts

Absorption of water and minerals –

Absorption of water and minerals from the soil by the roots and its ascent takes place by following processes-

1. Imbibition
2. Diffusion
3. Osmosis
4. Active Transport
5. Turgidity and Flaccidity (Plasmolysis)

1. Imbibition –

- Imbibition is defined as the phenomenon in which living or dead cells of plants absorb water by Surface Attraction due to hygroscopic constituents – cellulose, hemicellulose and pectin material of cell wall.

- Sometimes a closed can of dried seeds bursts open if water enters it. Because seeds absorb water by imbibition and swell up, which exerts a force on the walls of the can, and the can bursts open.

- Imbibition also helps in germination of seeds. When dry seeds are sowed in the soil, the dry dry contents of seeds - carbohydrates and proteins - imbibe water from their surroundings. When dry seeds swell up, the seed coat bursts to allow the embryo to come out.

2. Diffusion –

Diffusion is the movement of molecules of a substance (solid, liquid or gas) from the region of their higher concentration to the region of their lower concentration when the two are in a direct contact.

Significance of diffusion:

- Exchange of gases during respiration and photosynthesis takes place by diffusion.

- Loss of water in the form of water vapour from the leaves takes place by diffusion.
- Diffusion spreads molecules and ions throughout the protoplasm.

3. Osmosis –

Osmosis is the diffusion or movement of only solvent molecules from the region of their higher concentration to the region of their lower concentration through a semi-permeable membrane.

Inward and Outward Osmosis –

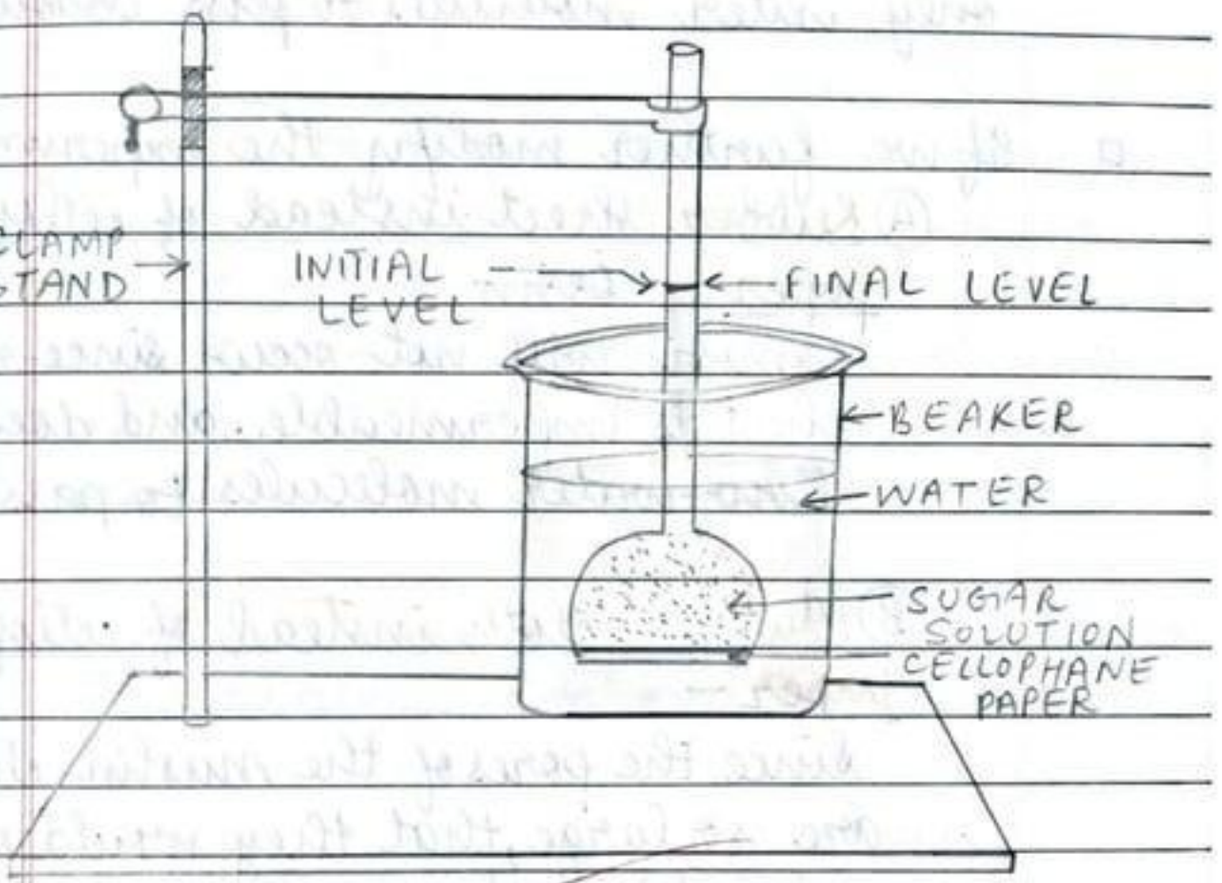
Direction of movement of molecules inwards or outwards with reference to a cell depends on the extent of concentration of solution surrounding it.

(a) ENDOSMOSIS (endo: inward)

It is the inward movement of water through a semi-permeable membrane when the solution surrounding ~~it~~ the cell is less concentrated. This tends to swell up the cell.

(b) EXOSMOSIS (exo: outward)

It is the outward movement of water through a semi-permeable membrane when the solution surrounding the cell is of higher concentration. This tends to <u>shrink the cell</u>.



- In the above experiment the cellophane paper acts as a selectively permeable membrane, water molecules are very small and sugar molecules are many times larger than water molecules.

- The pore of cellophane paper only allows water molecules to pass through but NOT sugar molecules.

- If we modify the experiment and use visking bag instead of cellophane paper the same observation will be recorded. This is because visking bag also acts as a semi-permeable membrane and allows only water molecules to pass through.

- If we further modify the experiment:
  - (A) Rubber sheet instead of cellophane paper – ~~Osmosis~~ osmosis will not occur since rubber sheet is impermeable and does not allow water molecules to pass through.

  - (B) Muslin cloth instead of cellophane paper –
    Since the pores of the muslin cloth are so large, that they would not not hold back even the sugar molecules, the ~~com~~ entire sugar solution flows down to a common level due to gravity. Muslin cloth is, therefore, freely permeable.

Significance of osmosis:

- Absorption of water from soil takes place through osmosis in root hair cells.

- Movement of water molecules from one

cell to another takes place through osmosis.

- Osmosis helps in maintaining the shape structure and form of different organs and parts of plant.

Osmotic Pressure – Osmotic pressure is the minimum pressure that must be created to prevent the passage of pure solvent into the solution when the two are separated by a semi-permeable membrane.

OR

Osmotic pressure is equal to the weight or pressure required to nullify osmosis.

Diffusion	Osmosis
Occurs in solids, liquids and gases.	Occurs only in solvent (liquid).
Diffusion does not require a semi-permeable membrane.	Osmosis requires a semi-permeable membrane.

TONICITY

The relative concentration of a solution that determines the inward or outward direction

of diffusion is called tonicity. Based on it, a solution can be of three types -

1. Isotonic
2. Hypotonic
3. Hypertonic

- **Isotonic Solution** - The relative concentration of water molecules and solute on both sides of the cell membrane is same. Therefore, there is no net movement of water molecules across the cell membrane.

- **Hypotonic Solution** - The solution outside the cell has lower concentration of solute as compared to the solution inside the cell. Therefore, water molecules move into the cell from the outside (ENDOSMOSIS).

- **Hypertonic Solution** - The solution outside the cell has higher solute concentration as compared to the fluids inside the cell. Therefore, water molecules move out of the cell from the inside (EXOSMOSIS)

4. Active Transport -

- Active transport is the passage of a substance from its lower to higher concentration through a living cell membrane

using energy from the cell.

- To obtain ions like Na^+ and K^+, they are forcibly carried inward from the region of lower concentration to the region of higher concentration with the help of special carrier proteins present in the membrane.

- Active transport is the opposite of diffusion; in active transport, movement is AGAINST concentration gradient whereas in passive ~~tan~~ transport, movement is TOWARDS concentration gradient.

5. Turgidity –

- When a cell is kept in water, it absorbs water through endosmosis and finally reaches a state where it cannot accomoda-te any more water. The cell is fully stretched and is called TURGID and the condition is called TURGIDITY

- The hydrostatic pressure of the cell sap on the cell wall is called turgor pressure.

- However the plant cell has a very strong cell wall, which reacts to the turgor pressure

and exerts pressure on the cell contents. This pressure is called wall pressure.

6. Plasmolysis –

- Plasmolysis is the contraction of cytoplasm from the cell wall caused due to the withdrawal of water when placed in a strong (hypertonic) solution.
- When the plasmolised cell is kept in normal water, protoplasm returns to its original position and is called deplasmolysis.
- The initial stage where plasmolysis just starts is called incipient plasmolysis.

7. Flaccidity –

- Flaccidity is the condition in which the cell content is shrunken and the cell is no more "tight".
- It is the ~~no~~ reverse of turgidity.

Importance of plasmolysis and turgidity

i) Plasmolysis is used to kill weeds in lawns and in preserving food items by salting

of bacteria.

ii) Turgidity is essential to maintain rigidity in leaves.

iii) Turgid cells provide mechanical support.

iv) The opening and closing of stomata depends upon the turgidity of guard cells.

v) Turgor pressure helps roots to crack the hard ground.

vi) In some plants, the movement of plant organs are caused due to the change in turgor pressure. for example: the drooping of the leaves of touch-me-not plant (Mimosa pudica), on touching them with finger, is due to the loss of turgor pressure at the base of the leaflets called PULVINUS.

Root Pressure

It is the pressure developed in the roots due to continued inward movement of water through cell-to-cell osmosis which helps in the ascent of cell sap

upward through the stem.

Some terms :

- **Bleeding or Exudation** — loss of water in the form of water droplets from the cut end of stem, due to root pressure.

- **Guttation** — Loss of water in the form of water droplets along the margin and tip of leaf, due to root pressure.

Ascent of Sap

The process of movement of sap from the roots to the aerial parts of the plant is called ascent of sap.

Factors contributing to ascent of sap –

1. Root Pressure – It builds up sufficient force to push the sap in the xylem vessels up to a certain height and may be enough for herbaceous plants.

2. Capillarity – (narrow diameter) of xylem vessels causes the water from a lower level to rise to fill the vacuum created by loss of water through transpiration

Narrower the diameter of a tube, greater will be the height of water rising in it and exerting a force called capillary force.

3. Transpiration Pull - As the water is lost through transpiration, a great ~~transpiration pull~~ water deficit takes place in cells. ~~This~~ This draws water osmotically from xylem vessels in leaf cells.

This pulling (suction) force provided by the leaves is specially important to produce a continuous column of water ☆ through stem.

☆ force of attraction between the water molecules is cohesive force.

4. Adhesion — It is an attracting force acting between two substances, such as water and the wall of xylem vessels. It causes the water to cling to the surface on which it rests

If the adhesive force between water and the cell wall is stronger, it will pull the water up to the top of the tree.

— × — × —

02/08/22

Ch- Genetics

Genetics is the study of heredity, i.e. the study of transmission of body features from parents to offsprings and the laws relating to such transmission.

There are two modern applications of genetics -

1) Genetic engineering - It is the technique in which the genetic constitution of an organism is altered by introducing new genes into its chromosomes.

e.g. production of genetically modified organisms such as crops.

artificial production of insulin hormone with the help of certain bacteria.

11) Genetic counselling - Newly married couples are advised to consult a specialist regarding the possibility of any undesirable trait which their children might inherit.

Genetic diseases which can be prevented by genetic matching of prospective parents.

e.g. Haemophilia (Bleeder's disease / Royal's blood disease)
Thalassemia
Sickle cell anaemia (RBC becomes sickle-shaped due to defective haemoglobin)

NOTE

Father of Genetics : Gregor Johann Mendel

Used the term 'Genetics' in 1905 : W. Bateson

Heredity

Heredity is the transmission of genetically based characters from parents to offspring.

It is the tendency of the offspring to inherit the characters of parents.

[Like begets like - It means that young ones look like their parents]

All animals produce their own kind but the offspring is never the exact copy of its parents. The offsprings show some differences in population, however small it may be.

Variation – The small differences among the individuals of the same species are called variations.

→ The greatest advantage of variation ~~is~~ in a species is that they increase the chances of survival in a changing environment.

→ All human beings are similar but they show so many races due to variations in their characters.

Heredity and Variation go side by side and are the basis of evolution.

SELECTION OF PEA PLANT BY MENDEL

Mendel conducted breeding experiments on garden pea (Pisum sativum).

The reasons for selecting garden pea for this experiment were –

i) Many varieties were available having contrasting characters.

ii) Varieties were available in pure form.

iii) Peas are self-pollinated since they are bisexual.

iv) Self-pollination could be prevented by removing corresponding reproductive parts of the flower.

v) Cross pollination could be done artificially

vi) Life cycle of a pea plant is very short.

vii) Pea plants produce large number of offsprings.

CHROMOSOMES

Carrier of hereditary units (genes).

Humans have 23 pairs of chromosomes

22 pairs (Autosomes)	1 pair (Sex chromosomes or Allosomes)
→ Determine the basic body features and characters of the individual.	→ Can be XX or XY which determines sex of the individual

Karyotype - The artificial arrangement of chromosomes according to their shape and size on a chart is called Karyotype.

Sex determination in Humans -

Female gamete (egg) contains only X chromosomes while in males, sperm are of two types - one bearing X-chromosomes and the other bearing Y-chromosome.

The sex of a child depends on the kind of chromosome carried by the sperm that fertilizes the egg.

During fertilization there are equal chances that an ovum gets fertilized either with X-chromosomal sperm or Y-chromosomal sperm.

↳ If an egg (X) fuses with X-chromosome bearing sperm, the resulting combination will be XX & ∴ the child will be a girl.

↳ If an egg (X) fuses with Y-chromosome bearing sperm, the resulting combination will be XY & the child will be a ~~son~~ boy.

GENES

Genes are specific sequences of nucleotides on a chromosome that encode particular proteins which express in the form of certain body feature.

Genes are the hereditary units responsible for transfer of characters from parents to offspring.

Genes work in pairs. Every gene has two alternative forms for a character producing different effects, these alternative forms of a gene are called ALLELES.

The genes are present in homologous pair for any particular trait.
Homologous chromosomes — They are a pair of corresponding chromosomes of the same shape and size, one from each parent. They bear same genes at same loci.

Trait - The alternative form of a character is known as trait.

Dominant gene	Recessive gene
The gene which expresses itself phenotypically in the presence of contrasting allele.	The gene which can ~~only~~ express itself only in homozygous condition.

Homozygous condition -
An individual in which both the alleles ~~are~~ of a gene are identical is called homozygous. e.g. TT (tall plant)

Heterozygous condition -
An individual in which the two alleles of a gene are different is called heterozygous condition for that character e.g., ~~Tt~~ Tt

Monohybrid cross - The cross between two parents having one pair of contrasting characters.

15.8.22

Ch- Photosynthesis

Photosynthesis is the physiological process by which plant cells containing chlorophyll produce food in the form of glucose or starch by using carbon dioxide, water and light energy. Oxygen is released as a by-product.

→ Only green plants prepare their food from simple inorganic molecules in the presence of solar energy. They are called autotrophs and their mode of nutrition is called autotrophic nutrition.

Chlorophyll- The Vital Plant Pigment

Chlorophyll is the green pigment found in green plants. It is found in cell organelles called chloroplasts.

It absorbs light of red and blue wavelength and reflects light of green wavelength. Therefore, a leaf appears green.

It is highly sensitive to light, but too much light may destroy it.

Chlorophyll a and chlorophyll b are most abundant in plant cells.

Chloroplast -

- The chloroplast is a membrane-bound, oval cell organelle.
- It is enclosed by a double membrane. Its interior contains closely packed flattened sacs called thylakoids. Chlorophyll is present in the walls of thylakoids. Thylakoids are the sites for light phase of photosynthesis.
- These thylakoids are arranged in piles called grana lying in a colourless ground substance called stroma. It contains ribosomes, circular naked DNA (DNA without histone proteins) and phospholipid globules and other organic substances.

  Dark reaction of photosynthesis takes place here.
- The grana are connected with each other by stroma lamellae or frets.
- There are about 40-50 chloroplasts

present in each leaf cell.

Raw materials for Photosynthesis:

Light, chlorophyll, carbon dioxide and water are essential requirements for photosynthesis.

Sunlight - Sun is the ultimate source of energy. In photosynthesis, light energy is converted to chemical energy.

Chlorophyll - Chlorophyll is the pigment located in the thylakoids of chloroplast which absorbs light energy.

Carbon dioxide - A source of carbon is needed for the synthesis of sugars.

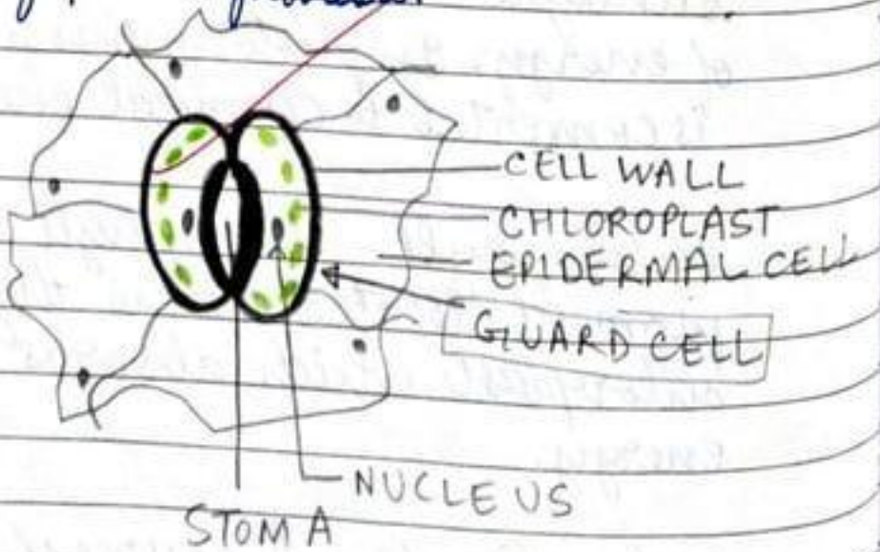
Carbon dioxide is reduced to form glucose.

Water- Hydrogen ions are needed for the synthesis of glucose, and water is the only source of hydrogen ions that plants can make use of, by splitting water into hydrogen and oxygen by utilising solar energy.

$$6CO_2 + 12H_2O \xrightarrow[\text{chlorophyll}]{\text{Sun's energy}} C_6H_{12}O_6 + 6H_2O + 6O_2\uparrow$$

Regulation of Stomatal Opening

Stomata are minute openings present in the epidermal layers of leaves. They are responsible for the exchange of gases during photosynthesis.



SHOT ON OPPO

K⁺ ion Concentration Theory -
The stomatal opening and closing depends on the generation of a potassium ion gradient.

- During day ATP produced during light reaction is used to actively pump potassium ions from the neighbouring cells into the guard cells, which increases the solute concentration inside the guard cell, thereby decreasing the water concentration of the guard cell.

- Water enters in the guard cells from the neighbouring / epidermal cells by simple process of osmosis. This causes opening of stomata.

- At night, the K^+ ions leak out, thus reducing the turgor of guard cells and the stomatal pore closes.

Process of Photosynthesis

The palisade layer is the centre for photosynthesis. Light energy is trapped in the chlorophyll of mesophyll cells of the palisade layer of leaves.

Two main phases are involved in the mechanism of photosynthesis:

LIGHT-DEPENDENT PHASE / PHOTOCHEMICAL PHASE

This phase takes place in thylakoids of chloroplas

Step-I) Activation of Chlorophyll: On exposure to light, chlorophyll becomes excited and absorbs light energy.

Step-II) Splitting of water: The energy absorbed is used in splitting of water molecules into hydrogen ion (H^+) and oxygen, releasing 2 electrons from each water molecule. This is called photolysis of water.

$$2H_2O \xrightarrow{\text{energy of 4 photons}} 4H^+ + 4e^- + O_2$$

End products of photolysis -

i) The hydrogen ions (H^+) are picked up by a compound NADP (Nicotinamide adenine dinucleotide phosphate) to form NADPH.

ii) The oxygen component (O) is given out as molecular oxygen (O_2)

$$2O \rightarrow O_2 \uparrow$$

iii) The electrons are used in converting ADP (Adenosine diphosphate) into energy-rich compound ATP (Adenosine triphosphate)

by adding one phosphate group (P_i (inorganic phosphate). This is called photophosphorylation

LIGHT INDEPENDENT PHASE / DARK PHASE / BIOSYNTHETIC PHASE

i) 'Dark reaction' does not mean that it takes place at night or darkness. It simply means that dark reaction does not require light energy.

ii) This stage occurs in the stroma of the chloroplast.

iii) Dark reaction involves the fixation and reduction of carbon dioxide, resulting in the formation of carbohydrates (glucose then starch), by utilizing ATP and NADPH obtained as a result of the light reaction.

Inter-conversion of Glucose and Starch

i) As soon as glucose is formed, several molecules of glucose are transformed to produce one starch molecule. This process is called polymerisation.

ii) At night, starch is reconverted into soluble glucose which is transported to other parts through phloem. This process of transporting sugar to different parts of the plant is known as translocation.

Jussmate
Date
Page

Adaptations in Leaf for Photosynthesis

Large Surface Area	Leaves are broad ~~to~~ for maximum light absorption
Leaf Arrangement	Leaves are at right angle to the light source to obtain maximum light.
Cuticle and upper epidermis	They are transparent and waterproof to allow light to enter freely.
Numerous stomata	A large number of stomata allow the rapid exchange of gases.
Thinness of leaves	It reduces the distance between cells facilitating rapid transport
Chloroplast	Chloroplasts are concentrated in the upper layers of the leaf.
Extensive vein system	An extensive vein system allows ~~for~~ rapid transport to and from mesophyll cells.

Factors Affecting Photosynthesis –

EXTERNAL FACTORS –

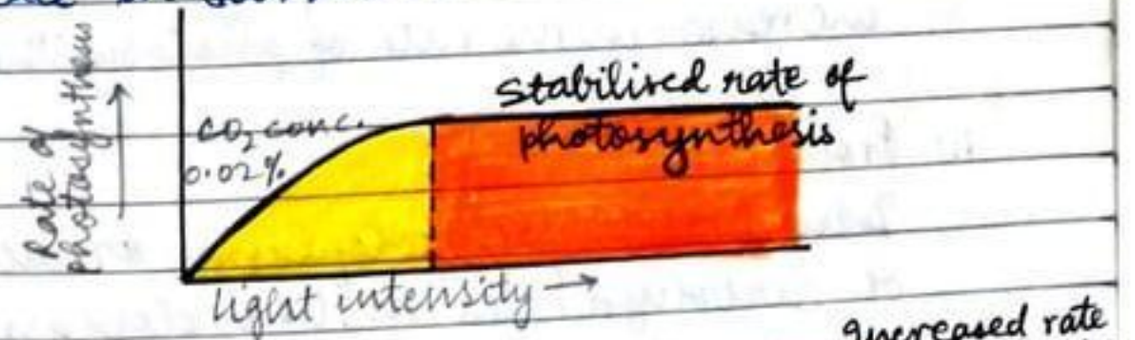
i) Light intensity and ii) Carbon dioxide

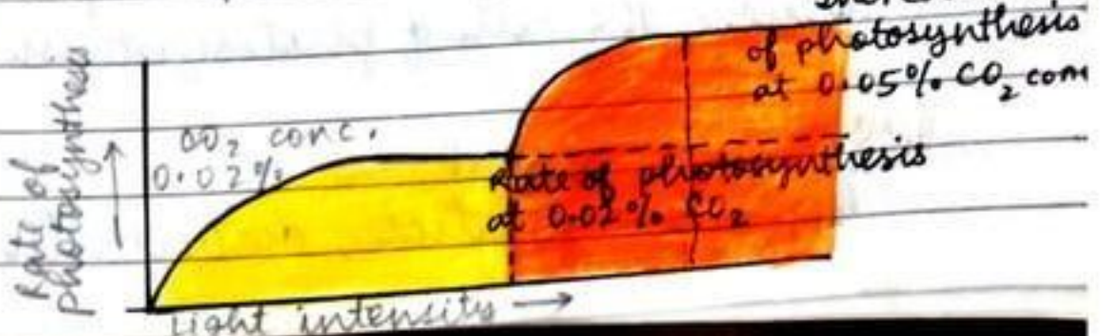
The rate of photosynthesis increases with light intensity up to a certain limit and then it stabilises at that point (at 0.02% CO_2), because CO_2 becomes a limiting factor.

An increase in CO_2 concentration at this point increases the rate of photosynthesis further, and again gets stabilised at a point (at 0.05% CO_2).

extremely intense light bleaches chlorophyll and retards photosynthesis.

Highest rate of photosynthesis occurs in Red and Blue light and the minimum in green light. No photosynthesis takes place in darkness.





iii) Temperature

The rate of photosynthesis increases with the rise in temperature, usually up to the optimum temperature 35°C. Above 35°C, rate of photosynthesis gradually decreases and stops above 40°C because enzymes become denatured above 40°C. Similarly low temperature also inhibits photosynthesis.

iv) Water Content

The scarcity of water due to excessive transpiration or reduced absorption by roots decreases the rate of photosynthesis.

INTERNAL FACTORS

i) Chlorophyll

More amount of chlorophyll results in increase in the rate of photosynthesis.

ii) Protoplasm

Dehydration of protoplasm or accumulation of carbohydrates in the protoplasm reduces the rate of photosynthesis.

iii) Structure of leaf

Thickness of cuticle, distribution of

stomata and the size of leaf influence the rate of photosynthesis.

Importance of Photosynthesis

i) Provides Food : All animals are directly or indirectly dependent on plants for their food. Hence photosynthesis is an important physiological process.

ii) Provides oxygen : Oxygen is present in a free state in the atmosphere because of photosynthesis. Almost all organisms need oxygen for respiration.

Carbon Cycle

The carbon cycle is a series of chemical reactions in which carbon as a chemical element (in CO_2) is removed from the air, used by living organisms in their body processes and is finally returned to the air.

Chapter - Transpiration

Transpiration is the process of loss of water in the form of water vapour from the leaves and other aerial parts of the plant.

Significance of Transpiration:

i) Cooling effect - Transpiration is useful to plants on hot sunny days because evaporation reduces temperature of leaf surface.

ii) Suction force - Transpiration helps in ascent of sap by producing a suction force acting from the top of a plant.

iii) Distribution of water and mineral salts - Transpiration from the surface of leaves tends to draw water ~~towards~~ upwards and thus helps in the distribution of water throughout plant body.

iv) Climate - Transpiration increases moisture in the atmosphere and brings ~~rain~~ rain. Thus it regulates water cycle.

Measurement of Transpiration : By Potometer

A potometer is a device which measures the rate of water intake by a plant. This water intake is almost equal to the water lost through transpiration.

Limitations of Ganong's Potometer :

i) Introducing the air bubble is not very easy.

ii) The twig may not remain fully alive for a long time.

iii) Any change in the outside air temperature may affect the position of the air bubble in the capillary tube.

Precautions in the use of potometer :

i) The potometer should be made completely water-tight.

ii) The twig should be cut obliquely and under water to avoid suction of an air bubble into the twig, which will stop the absorption of water into the xylem.

Kinds of Transpiration

i) Stomatal transpiration – From leaves through stomata
ii) Cuticular transpiration – Directly from the surface of leaves and stems
iii) Lenticular transpiration – From the lenticels which are the minute openings on the surface of old woody stems.

Mechanism of stomatal transpiration:

i) Most of the transpiration takes place through stomata. The number, structure, distribution, opening and closing of stomata influence the rate of transpiration.

Almost 90% of the total water loss from a plant takes place through stomata.

ii) As water molecules are lost by evaporation and moved out of the xylem, the whole column of water in the xylem is pulled upwards due to cohesion (force of attraction between same molecules) between water molecules.

iii) Water is moved across the root by osmosis to maintain continuous column in the

xylem.

iv) The water absorbed by root hairs enters xylem vessels of the root and rises up to reach the xylem vessels of the stem.

Finally it enters the vessels of fine veins of leaves throughout the midrib.

v) A large number of spongy mesophyll cells in the leaf have their surfaces exposed to the intercellular spaces.

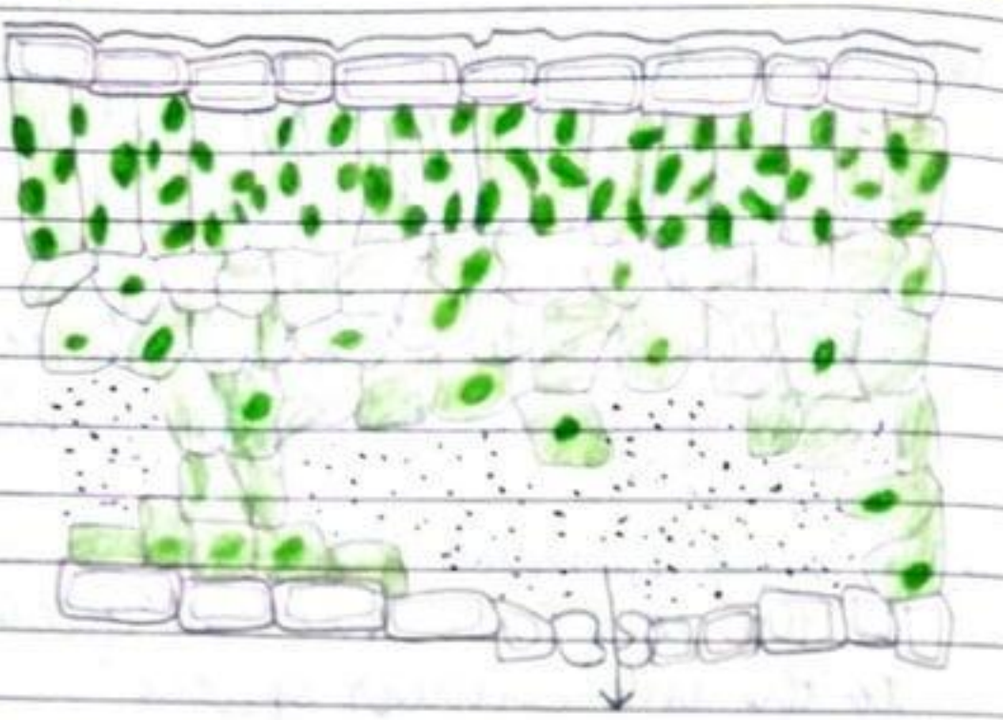
vi) The surfaces of the cells give out some of the water which forms a thin film.

vii) The water from this film evaporates and the water vapour formed saturates in the intercellular spaces.

viii) This vapour then diffuses into the other connecting intercellular spaces and finally reaches substomatal space, from where it escapes out through the stomata.

ix) Stomata are present more on the lower surface of the leaves in dicots (rose, banyan and mango) and on both

surfaces of monocots (maize, grass).



Factors affecting transpiration-

Ⓐ INTERNAL -

i) Leaf area - Larger surface area of leaves means more transpiration

ii) Leaf structure - Presence of thick cuticle, coating of wax, sunken stomata, compact mesophyll, reduction in number of stomata REDUCE the rate of transpiration.

iii) Rolling, twisting, curling - It reduces the rate of transpiration because the amount of radiation received by the leaf is decreased.

iv) Mucilage or latex - Reduces the rate of transpiration.

v) Root-shoot ratio - If the root system of a plant is more efficient, the rate of transpiration is more.

(B) EXTERNAL/ENVIRONMENTAL -

i) Light - During day the stomata opens due to synthesis of glucose in the guard cells. More transpiration occurs during the mid-day.

On a cloudy day, stomata gets partially closed thus reduce transpiration.

ii) Temperature - Increase in temperature increases transpiration (upto a certain limit) as warm air can hold more water than cold air.

iii) Wind - Fast blowing wind removes the water vapour from the surface of the leaves, thus increasing transpiration.

iv) Humidity - If the atmosphere is more humid, transpiration reduces as the outward diffusion of water reduces.

v) Water content of soil – Lesser is the water content of soil, lesser is the the transpiration. At the state of lesser water content, the leaves wilt and the stomata gets closed. So transpiration reduces.

vi) CO_2 concentration – If atmospheric CO_2 increases above normal (0.03%) then stomata gets closed and the transpiration reduces.

vii) Atmospheric pressure – When atmospheric pressure increases, transpiration rate decreases and vice-versa.

Methods to Reduce Excessive Transpiration

i) Sunken stomata – It is observed in xerophytes or desert plants eg, Nerium (oleander), # In some plants, stomata are covered by hair to reduce transpiration.

ii) lesser number of stomata – It reduces transpiration

iii) Leaves turn wavy rolled or folded – This

reduces surface area and thus transpiration as well.

iv) Narrow leaves – Plants like oleander have narrow leaves that reduce the surface area and thus transpiration.

v) Loss of leaves - In some cases, leaves may get be dropped or may be absent or changed into spines as in most cacti.

vi) Thick cuticle – The leaves may be covered by thick cuticle, e.g., Banyan, and most evergreen trees.

Other ~~ways~~ types of transpiration:

② Cuticular Transpiration

Cuticle is a waxy layer secreted by the epidermis on the two surfaces of the leaf. The thickness of the cuticle varies from plant to plant. Primarily, the cuticle serves to prevent evaporation of water from the leaf surfaces. However, some evaporation does occur.

The greater the thickness of the cuticle, the lesser is the evaporation (transpiration). Desert plants tend to have thicker cuticles to cut down transpiration.

③ Lenticular Transpiration

Lenticels are special openings that develop on the barks of older stems in place of stomata. These allow diffusion of gases for respiration as well as for photosynthesis.

Lenticels never close. They remain open all the time. Water from the cell surfaces directly facing the lenticel evaporates and contributes to transpiration.

The amount of transpiration from lenticels is certainly more than the cuticular transpiration, but very much less than the stomatal transpiration.

Guttation –

It is the exudation of water drops from the margins or tips of leaves through special openings called hydathodes.

It occurs due to high root pressure and low temperature. ex - oats, tomatoes, potatoes, grass, sunflower.

Bleeding –

It is the exudation of sap from injured parts of plants. It is also a manifestation

of root pressure. If an ~~was~~ incision is made in the stem of a plant growing in well-watered soil, xylem sap starts oozing out due to root pressure.

Ex - Exudation of milky latex in Calotropis and Euphorbia.

Evaporation	Transpiration
Loss of water from the surface of water bodies (in the form of vapour).	Loss of water (in the form of vapour) from aerial parts of the plant.
It is controlled by the temperature and humidity of the atmosphere.	It is controlled by both internal and external factors.
It is a fast process	It is a slow process
Transpiration	**Guttation**
Water is lost in the form of vapour.	Water is lost in the form of water droplets.
Water vapour is lost through stomata, lenticels and cuticle.	Water is lost through pores (hydathodes) found at the ends of veins.
It occurs during dry conditions.	It occurs during humid conditions.

29.9.22

Chapter - Circulatory System

Types of circulatory system -

1. Open circulatory system The blood flows through open spaces in the body.

2. Closed circulatory system - The blood flows through vessels.

Blood :

- Opaque viscous liquid
- Bright red in artery (oxygenated) and dark red in veins (deoxygenated).
- 5 to 6 litres by volume
- Slightly alkaline (pH 7.3 to 7.45)

Functions -

i) Helps in transportation of respiratory gases, excretory material and hormones. It also distributes heat.

ii) WBC in the blood provides protection by destroying the germs by phagocytosis and by producing antibodies.

iii) The platelets in the blood help in clotting which prevents excess bleeding and also

entry of germs into the body.

BLOOD

- Plasma
  - Water (90-93%)
  - Proteins (7-8%) (fibrinogen)
  - Inorganic salts (9%) (NaCl, Sodium bicarbonate)
  - Other substances (Glucose, hormones, amino acids)
- Cellular elements
  - RBC (erythrocytes)
  - WBC (leukocytes)
  - Blood platelets (thrombocytes)

The plasma from which the protein fibrinogen has been removed is called serum.

Red Blood Cells (Erythrocytes) –

i) Biconcave disk-like structures
ii) 7 μ (micron) in diameter.

Adult ~~male~~ man: 5 million RBC's per mm of blood

Adult woman: 4.5 million RBC's per mm of blood

What is the advantage of small size of RBCs?
The small size and concavities on either side of the RBCs provides a large surface area which makes them very efficient in absorbing oxygen.

Also their small size enables them to travel through very fine capillaries in the body where they have to travel in a single file.

Iron pigment in blood - Haemoglobin

Life and Death of RBCs -

Produced in -

- In adults - bone marrow of long bones
- In children - bone marrow of all bones
- In embryo - in liver and spleen

Average life - 120 days

Destroyed in - Spleen, liver and bone marrow

NOTE : i) Mature RBCs have NO NUCLEUS! This makes the cell biconcave and thus increasing their surface area to volume ratio for absorbing more oxygen.

ii) No mitochondria - The Red Blood Cells

cannot use oxygen for themselves. Thus all oxygen absorbed from the lungs is transported and delivered to the tissues unconsumed. Secondly ~~less no. of~~ loss of mitochondria means full transport of glucose in blood plasma, unused by the RBCs.

iii) No endoplasmic reticulum – Means increased flexibility of RBCs for their movement through narrow capillaries.

Abnormally increased no. of RBCs – Polycythaemia

Abnormally decreased no. of RBCs – Erythropenia

White Blood Cells (Leucocytes) –

i) Contains a nucleus
ii) Does not contain haemoglobin
iii) 4000 to 8000 WBCs per mm^3 of blood
iv) Produced in red bone marrow and lymph nodes
v) life span: two weeks

Leukemia – A cancer of tissue forming WBCs, whose number increases manifold at the cost of RBCs.

classmate
Date
Page

Leukopenia – Abnormal decrease in the number of WBCs

Functions of WBCs –

i) Phagocytosis – This is a process in which most WBCs and particularly the neutrophils engulf particles like solid substances, especially bacteria and act as a defense mechanism against disease-causing germs.

ii) Inflammation – Occurs due to the reaction of tissues to injury and to localised invasion of germs. Here the leucocytes (specially monocytes and neutrophils) migrate through the wall of blood vessels by diapedesis and fight against disease-causing germs.

iii) Formation of antibodies – The WBCs (specially lymphocytes) produce antibodies which kill or neutralise the germs or the poisons from them.

WHITE BLOOD CELLS

Granulocytes

- cytoplasm contains granules
- Nucleus usually constricted into lobes

→ Neutrophils (62%)

i) Nucleus with 3-4 lobes
ii) Granular cytoplasm
iii) Engulfs bacteria by phagocytosis
iv) Produced in bone marrow

→ Eosinophils (~~2-3%~~ 2.3%)

[Its count increases in allergies]

i) Nucleus with 2 lobes
ii) large cytoplasmic granules
iii) Engulf bacteria
iv) Secrete antitoxins
v) Associated with allergy
vi) Produced in bone marrow

Agranulocytes

- Cytoplasm does not contain granules
- A single large nucleus

→ Lymphocytes (30%)

i) Smallest of WBCs
ii) Single large nucleus
iii) Produces antibodies
iv) Produced in bone marrow and lymph gland.

→ Monocytes (5.3%)

i) Largest of WBC
ii) Nucleus is large and kidney-horse shoe shaped
iii) At the site of infection ~~they~~ they transform into macrophages
iv) Ingest germs
v) Produced in bone marrow.

(Contd.)

→ Basophils (0.4%)

i) Nucleus large and indistinctly lobed

ii) Release chemicals (histamine) for inflammation which dilate blood vessels

RBC (erythrocytes)	WBC (leukocytes)
i) Do not have a nucleus	Have a nucleus
ii) Contain haemoglobin.	Do not contain haemoglobin
iii) In male: 5million/mm^3 In women: 4.5million/mm^3	4000-8000 per mm^3 of blood
iv) Produced in the marrow of long bones in adult and bone marrow of all bones in children.	Produced in red bone marrow, lymph nodes and sometimes in the liver and the spleen.
v) life span - 120 days	v) life span - 2 weeks
Function - It has an iron pigment called haemoglobin which is located in stroma of RBC.	Function - They destroy germs by phagocytosis. They produce antibodies (lymphocytes) They are

Haemoglobin has strong affinity for oxygen. ~~Oxy~~ oxyhaemoglobin Hence RBCs help in transportation of oxygen as oxyhaemoglobin and it is transports some CO_2 as carbamino-haemoglobin	also responsible for inflammation at the site of injury of infection
Abnormal increase in number of RBCs- Polycythaemia	Abnormal increase in WBCs- Leukemia
Abnormal decrease in number of RBCs- erythropenia	Abnormal decrease in WBCs- Leukopenia

Blood Platelets -

i) oval or round structure
ii) non-nucleated
iii) number : 2,00,000 to 4,00,000 per mm^3 of blood
iv) Formation : by breakdown of giant bone marrow cells called Megakaryocytes
v) Life span : 3 to 5 days

vi) Place of destruction - spleen

vii) Function - Helps in clotting of blood by secreting a chemical substance called thrombokinase. They disintegrate at the site of injury to secrete thrombokinase which initiates the process of clotting.

Blood Clotting

i) At the site of injury as the platelets flow over injured rough surface, they disintegrate to release a chemical substance called thrombokinase (also called thromboplastin)

ii) Thrombokinase converts inactive prothrombin into active thrombin in the presence of calcium ions.

iii) Thrombin then converts soluble fibrinogen into insoluble fibrin.

iv) The fibrin forms a network at the site of injury which traps the blood corpuscles. Then it shrinks and solidifies to form clot or thrombus.

v) At the site of clotting a pale yellow coloured liquid gets collected called serum.

Blood serum - It is the clear liquid part of the blood lacking red blood cells and protein fibrinogen.

Why blood doesn't clot inside the blood vessel?
The protein essential for clotting - prothrombin and fibrinogen are present in inactive form.

There is an anticoagulant called heparin

A rough surface is required for the platelets to disintegrate to release it thrombokinase, which initiates the process of clotting. But the inner lining of blood vessel is formed of smooth endothelium i.e., squamous epithelium which prevents platelets to disintegrate.

Diseases related to Platelets -

Haemophilia - It is a genetic disorder in which blood fails to clot and this is caused due to absence of clotting factor - prothrombin or fibrinogen.

Viral dengue - The number of platelets in the blood decreases.

Important - fat
Vitamin K, a soluble vitamin is essential for the production of prothrombin.

Blood Groups -

Universal Donor → O
Universal Recipient → AB

Rh factor - Rh stands for Rhesus, our common monkey.

BLOOD CIRCULATORY SYSTEM

The Heart -

Location - located right in the centre of the body, between the two lungs and above the diaphragm.

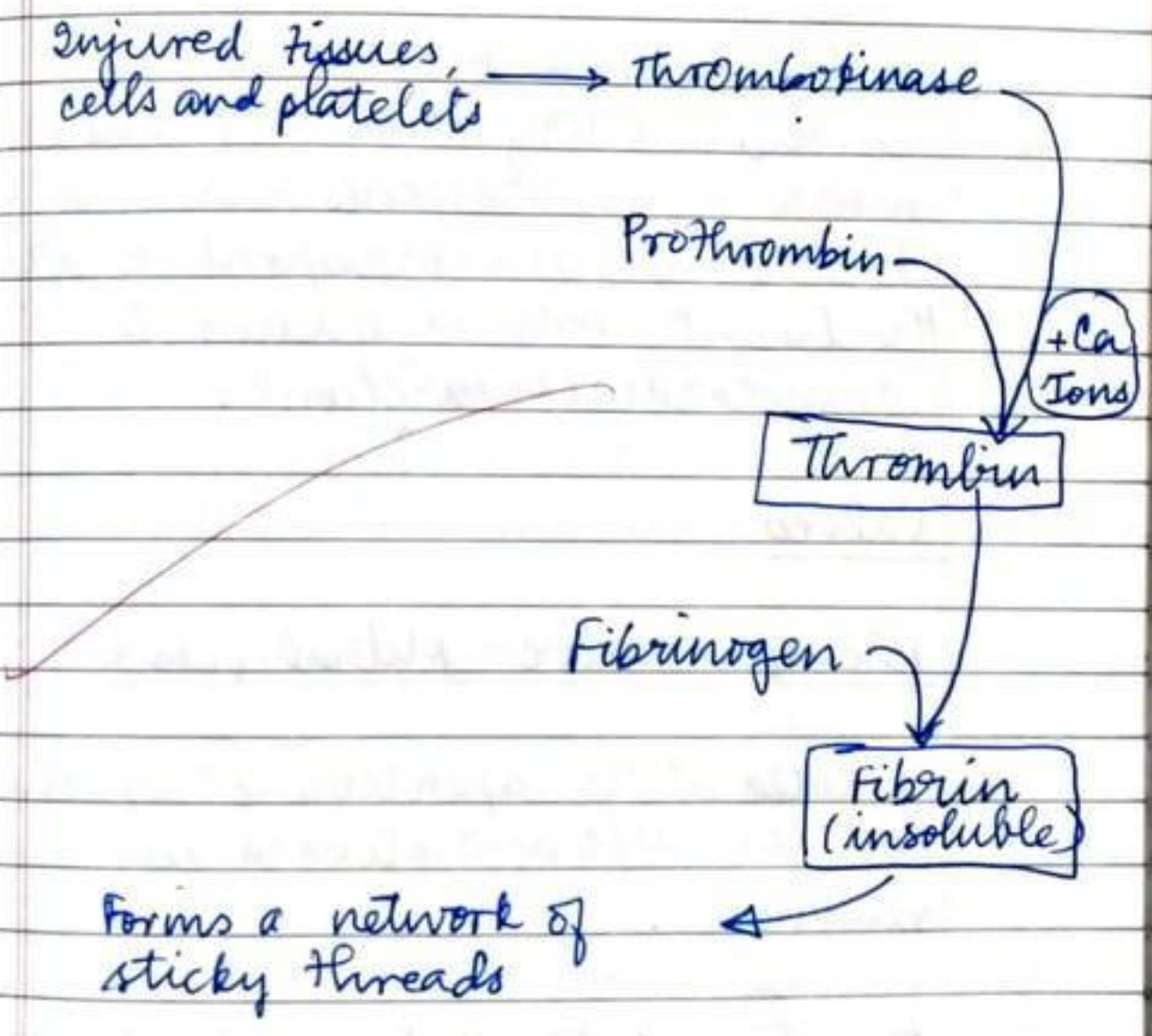
The heart is protected by a double walled membranous covering called pericardium. It contains lubricating pericardial fluid which reduces friction during heartbeat and protects it from mechanical injuries

The auricles have thinner walls because

their major function is to receive blood from the body and pump it into the very next ventricles.

The ventricles have thick muscular walls because they have to pump to the farthest points in the body.

BLOOD CLOTTING



12.10.22

Coronary arteries -

Location - A pair of coronary arteries arises from the base of aorta and branches and rebranches in wall of the heart.

Function - It supplies oxygenated blood to the walls or muscles of the heart.

When there is a blockage in any coronary artery or in any one or more of their branches, there is a deadening of the corresponding area of the heart muscles leading to 'myocardial infarction".

Valves -

Bicuspid Valve - Mitral valve

Located at the aperture or opening between left auricle and left ventricle.

Function - Prevents the backflow of blood from left ventricle to left auricle during ventricular systole.

Tricuspid Valve

located at the aperture between right auricle and right ventricle.

Function – It prevents the backflow of blood from right ventricle to the right auricle during ventricular systole.

Pulmonary Semilunar Valves –

located at the origin of pulmonary artery from right ventricle

Function – To prevent backflow of blood from pulmonary artery into right ventricle during ventricular diastole

Aortic Semilunar Valves –

located at the point of origin of aorta from the left ventricle.

Function – To prevent backflow of blood from aorta to left ventricle during ventricular diastole.

Chordae Tendinae

location - They are tendinous threads located in the ventricles extending between the cuspid valves and papillary muscles.

Function - They hold the cuspid valves in position and prevent them from over turning into the auricles during the ventricular systole

Heart Beat (Cardiac Cycle)

Contraction of atria = 0.15 s

↓

Contraction of ventricles = 0.30 s

↓

Joint diastole = 0.40 s

Sounds of Heart - The first LUBB is produced when the atrio ventricular valves get closed sharply at the start of ventricular systole.

The sound 'DUP' is produced at the beginning of ventricular diastole, when the semilunar valves at the roots of aorta and pulmonary artery get closed.

Pacemaker - (SAN)
- Located in the walls of the heart
- Function is to control the contraction and relaxation of heart chambers.

Atrio ventricular node (AVN) -
- It is a specialised muscle fibre located in the wall of right auricle close to the inter-auricular septum above the tricuspid valve.

- Function is to transmit the impulses from SAN to ventricular walls through bundle of HIS and Purkinje fibres to initiate ventricular systole.

Conducting system :

SAN → AVN → Bundle of HIS → Purkinje fibres → Ventricular walls

Smaller the size of the body, faster is the heartbeat. Give reason.
This is because smaller the animal, the more it loses its body heat due to higher surface area to volume ratio. Therefore the increased heart rate distributes body heat.

Important - The young growing stages such as human baby have higher metabolism for body growth and therefore the faster heart rate keeps the supply and 'take off' of the metabolic substances in right quantities.

Blood Vessels -

① Artery:

It carries blood away from the heart towards any organ.

Has thick muscular walls and narrow lumen.

The blood in it flows in spurts which correspond to ventricular systole of heart.

② Capillary:

Its wall consists of a single layer of squamous epithelial cells (endothelium). →

It has no muscles.

Function - To allow outward diffusion of

oxygen into the intercellular fluid, & from there into the tissue cells.

ii) To allow leukocytes to squeeze out through the capillary walls by diapedesis (amoeboid movement)

Complete the left over part

Absent
Absent work
17/10/20

The Two Blood Circulations — PULMONARY & SYSTEMIC

Blood circulation in the human body is called "double circulation" because blood flows twice in the heart before it completes one full round.

1) Pulmonary Circulation - Right ventricle → Pulmonary artery → Lungs → Pulmonary vein → Left auricle.

2) Systemic Circulation - Left ventricle → Aorta → Arteries → Various body parts & tissues → Veins → Superior & Inferior Vena Cava → Right auricle

HEPATIC PORTAL SYSTEM

A portal vein is one which starts with capillaries and also ends in capillaries.

The veins starting from stomach and intestines enter the liver as a combined hepatic portal vein.

There it breaks up into capillaries and a new vein (hepatic vein) is formed by their reunion. This hepatic vein joins the posterior vena cava.



Utility of Hepatic Portal System -

The food absorbed by the stomach and intestines is first brought to the liver which acts like a store to regulate the quantity of nutrients flowing into the general ~~blod~~ blood circulation.

Also if ~~poisons~~ a certain poisons are absorbed through food, they are detoxified in the liver.

The Pulse – Pulse is the alternate expansion and elastic recoil of the wall of the artery during ventricular systole.

Blood Pressure – It is the pressure which the blood flowing through the arteries exert on their walls.

Blood pressure can be measured with the help of sphygmomanometer.

★ Normal Blood Pressure: Systolic – 100-140mm
Diastolic – 60-80 mm

A rise in blood pressure above 140/90 is known as hypertension.

TISSUE FLUID AND LYMPH

As the blood flows in the capillaries of the tissues, the plasma and leucocytes leak out through their walls. This fluid bathes the cell and is called TISSUE ~~FUU~~ FLUID.

Functions – The oxygen and other substances (present in tissue fluid) are absorbed by the cells and CO_2 and other wastes is given out into tissue fluid.

Lymph – It is straw-coloured fluid which filters out of capillary walls and enters the lymphatic system. It does not have RBCs and platelets.

Functions of Lymph –

i) To supply nutrition and oxygen to those parts where blood cannot reach

ii) To absorb fats from small intestine through lacteals

iii) Lymphocytes and monocytes of the lymph function to defend the body ~~from germs~~

iv) **Function of Lymph Nodes –**
They tend to localize the infection and prevent it from spreading to the body as a whole.

The Spleen –
Location: Reddish-brown organ located in the abdomen behind the stomach and above the left kidney.

Function: i) ~~It do~~ To produce lymphocytes.
ii) To destroy worn-out RBCs.
iii) In embryo, spleen produces RBCs.

iv) To act like a blood reservoir and to release the stored blood into blood stream in an emergency (like haemorr-hage)

Pus: dead WBCs and tissue cells destroyed by bacteria form pus at the site of inflammation

The Excretory System

Excretion -
It is the process of removal of harmful and unwanted nitrogenous waste materials from the body.

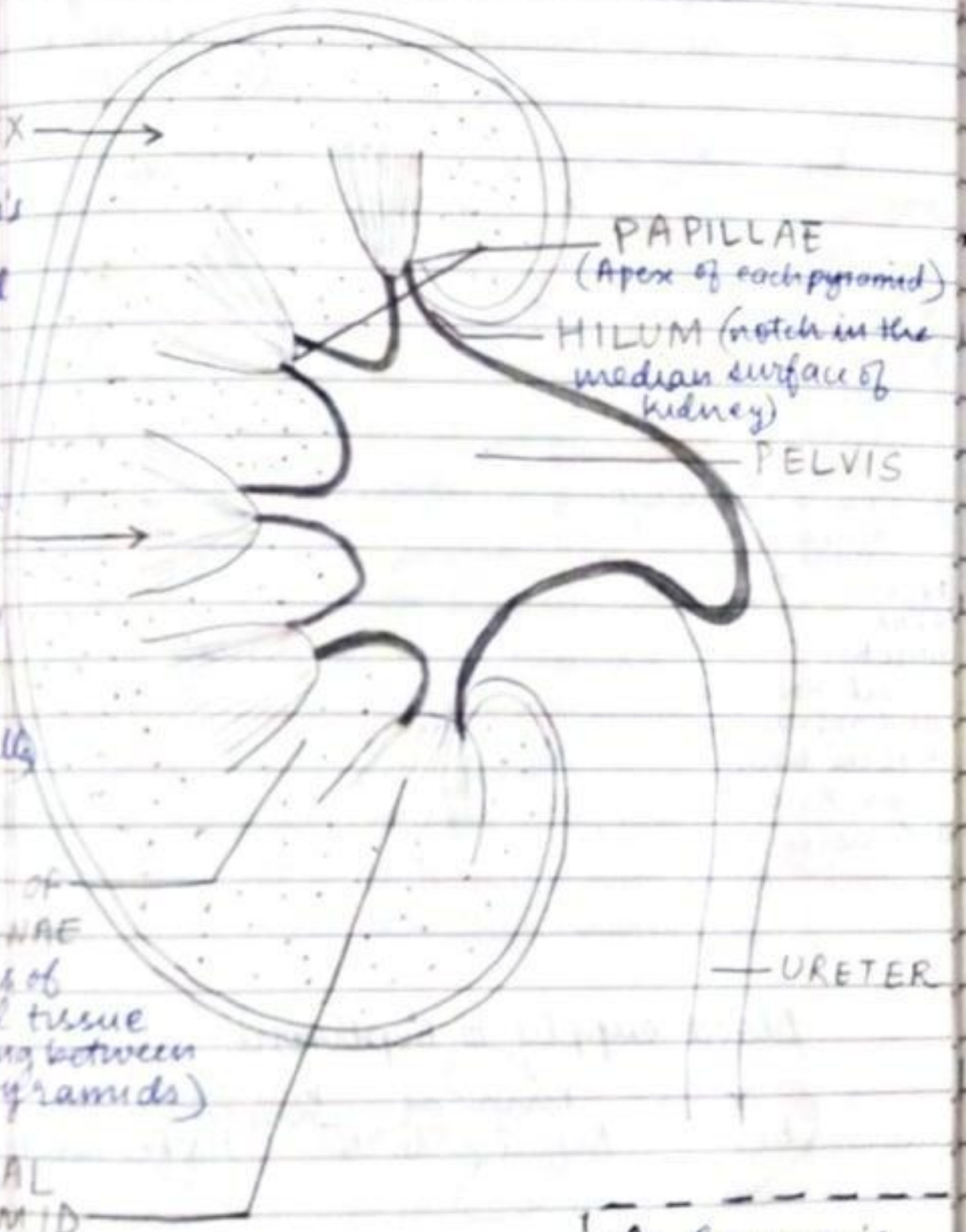
Kidneys are the primary excretory organs which eliminate urea and other nitrogenous wastes from blood.

Accessory excretory organs -
- Skin (secretes water and sodium chloride)
- Lungs (pass out carbon dioxide)
- Liver (detoxifies ammonia by converting it to urea)

KIDNEYS

- Bean shaped organs
- Located on either side of backbone and protected by the last two ribs.
- Right kidney is at slightly lower level than left one because the liver is present on the right side.
- Ureter arises from the notch (hilum) in the median surface of each kidney and and connects behind with the urinary bladder.

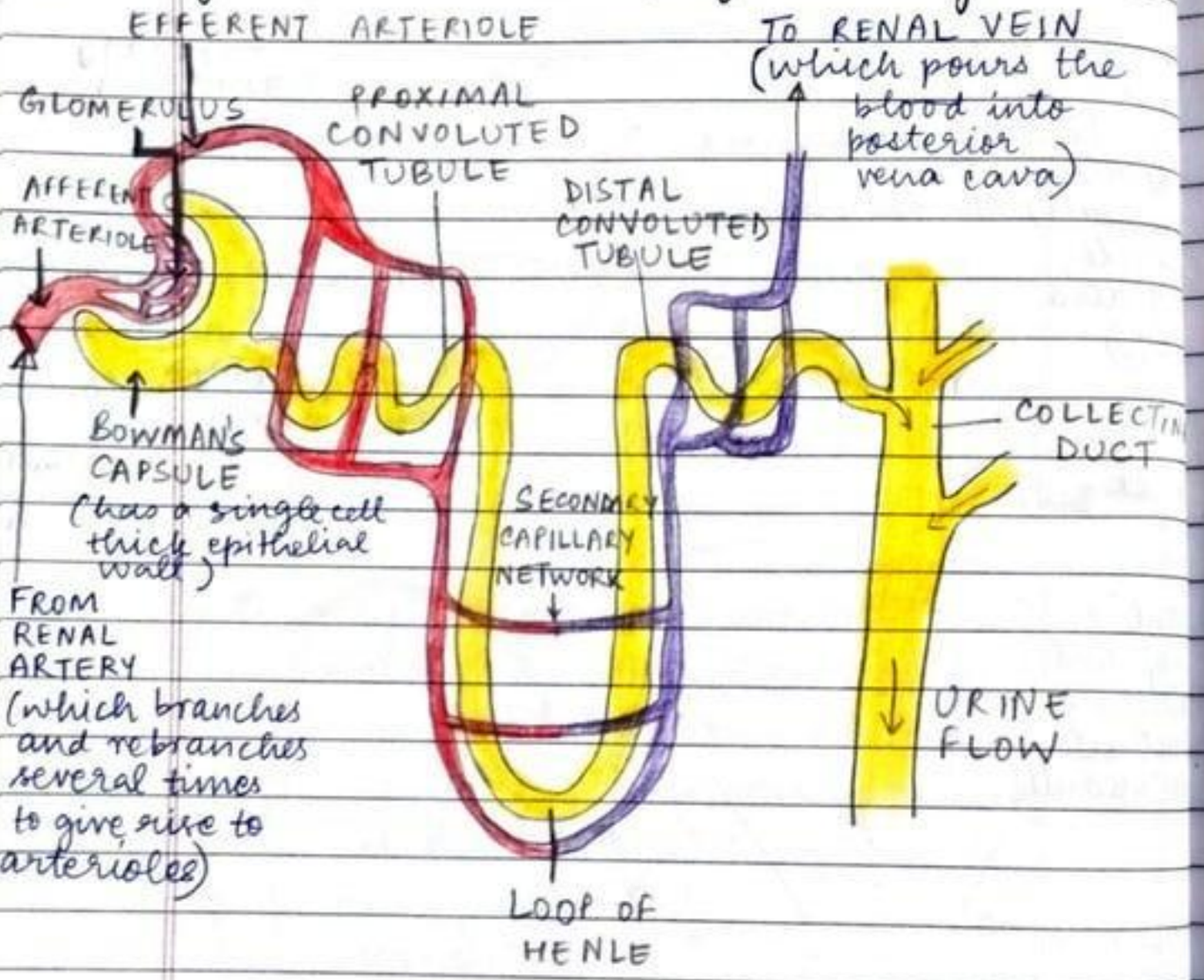
Longitudinal section of a kidney



RENAL PYRAMID
(structures composed of ... tubules arranged in ... conical pyramids)

★ Bowman's capsule and PCT give dotted appearance to Renal cortex in sectional view

Nephrons – The ~~structural~~ and functional unit of Kidneys :



Blood supply to Nephrons

Aorta → Pair of Renal arteries → (branches & rebranches) Afferent Arteriole

Afferent Arteriole → (form knot like mass (glomerulus)) Efferent Arteriole

Efferent Arteriole → (breaks up into) Secondary Capillary Network (Vasa Recta)

Secondary Capillary Network → (Rejoins to form) Renal vein → Posterior Vena Cava

PRODUCTION OF URINE

① Ultrafiltration

- Efferent arteriole is narrower than afferent arteriole, so blood flows through glomerulus under great pressure.

- The high hydrostatic pressure causes the blood plasma to filter out into the Bowman's Capsule. (Ultrafiltration)

- This fluid has water, urea, salts, glucose etc and is called glomerular filtrate.

- Blood in efferent arteriole is relatively thicker because most plasma is filtered out and only blood cells, proteins and other large molecules are left behind.

② Selective Reabsorption

- In PCT, two-thirds water and much of glucose, sodium and chloride ions are reabsorbed - only to the extent that the normal concentration of the blood is not disturbed.

- In Loop of Henle some water and Na^+ ions are reabsorbed. In DCT, some water and

chloride ions are reabsorbed.

(3) Tubular Secretion

- The walls of DCT secrete potassium and foreign chemicals such as penicillin into the forming urine

- The filtrate left after reabsorption and tubular secretion is called urine.

MICTURITION -
The process of expelling urine from the urinary bladder through the urethra is called micturition.

It occurs by the relaxation of sphincter muscles located at the opening of urinary bladder, under impulse from the brain.

Regulation of Urine Output -
Concentration of urine by water reabsorption is controlled by ADH (Antidiuretic hormone) secreted by posterior lobe of pituitary gland.

Tea, coffee, alcohol etc are called diuretics because they increase the formation of urine.

Abnormal Conditions Related to Urine

1. Haematuria – Blood passes with urine due to infection in urinary tract, kidney stones or tumour.

2. Glycosuria – Excess glucoses passes with urine due to diabetes mellitus

3. Albuminuria – Albumin proteins pass with urine due to high b.p. or bacterial infection. (Bowman's Capsule membrane becomes more permeable)

4. Bile pigments in urine – Due to jaundice, anaemia or liver cirrhosis.

5. Kidney stones – Formed by precipitation of uric acid and salts like calcium oxalate when the kidneys are not working normally

Physical Properties of Urine

- Yellow colour due to urochrome pigment
- 1 to 1.5 litres per day but varies
- Slightly acidic (pH = 6) usually
- Saltish in taste
- 95% water and 5% dissolved solid wastes ex- Urea, Creatinine, Sodium chloride, Potassium chloride, Ammonia, Uric acid
- Faint smell (but on standing, smell becomes strong)

Define –

① Osmoregulation – Regulation of osmotic pressure of blood by maintaining salt and water balance is called osmoregulation.

② Artificial kidney – When both kidneys of a person stop working, an artificial kidney i.e., a dialysis machine is used to remove nitrogeneous wastes and excess salts from blood.

③ Gout – A health condition in which uric acid crystallizes and gets deposited in the joints.

④ glomerular filtrate – The fluid entering renal tubule from the glomerulus is called glomerular filtrate. It is composed of water, urea, salts, glucose etc.

THE ENDOCRINE SYSTEM

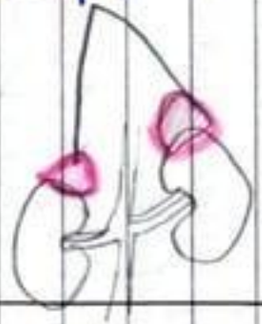
Endocrine glands are ductless glands which pour their secretions directly into the blood.

Hormones are chemical messengers secreted by endocrine glands directly into the blood. They influence a specific organ (target organ) and help in chemical coordination of the body.

Properties of Hormones –

i) secreted directly into the blood
ii) influence a specific organ (target organ)
iii) produced in very small quantities and biologically very active
iv) not stored in the body and excreted
v) chemically, hormones can be –
- ~~proteins~~ peptides – insulin
- amines – adrenaline
- steroids – testosterone

Endocrine Glands	v/s	Exocrine Glands
i) Ductless glands		They have duct present.
ii) Secretions are hormones		Secretions are enzymes, sweat or tear
ex- Adrenal gland, Thyroid glands		Sweat gland, Salivary glands.

Name of Gland	Exact location	Secretions
Adrenal gland	~~cap like glands~~ located on the top of each kidney like caps 	Adrenaline, Noradrenaline — By adrenal medulla cortisone and other cortical hormones like Mineralocorticoids, Glucocorticoids, Sex corticoids — By adrenal cortex
Pancreas 	Heterocrine gland located behind the stomach in the upper-left abdomen.	Insulin — by beta cells Glucagon — by alpha cells
Thyroid 	Butterfly-shaped gland situated in front of the neck just below the larynx.	Thyroxine

CASSMATE Date Page

Pituitary



"master gland" because its secretions control the secretions of almost all other endocrine glands.

Pea-shaped gland which hangs from the base of the mid-brain below the hypothalamus.

TROPIC Hormones:
The hormones which stimulate other endocrine glands to produce certain hormones.

TROPIC HORMONES

Posterior lobe secretes –
- Oxytocin
- Vasopressin (ADH)

Anterior lobe secretes –
- Growth Hormone
- Thyroid Stimulating Hormone
- Adrenocorticotropic Hormone
- Gonadotropic Hormones
  - Follicle stimulating hormone
  - Luteinizing hormone
  - Prolactin

HORMONES

Endocrine Gland	Hormones Secreted	Function
ADRENAL GLAND		
i) Adrenal medulla	Adrenaline	To prepare the body for emergenc heartbeat, faster and deeper bre eye.
ii) Adrenal Cortex	Mineralocorticoids	Regulates plasma sodium and concentration
	Glucocorticoids	Regulates carbohydrate, fat metabolism
	Sex corticoids	Stimulate development of sex. males
	Cortisone	Suppresses inflammation
PANCREAS		
i) Beta cells	Insulin	It checks the rise of sugar level in blood
ii) Alpha cells	Glucagon	It raises the sugar level in blood by breakdown of glycogen into glucos
THYROID GLAND	Thyroxine	It regulates basal metabolism. It growth of the body, mental ~~developed~~ ossification of bones.

AT A GLANCE

	Undersecretion	Oversecretion
Causes faster & dilates pupil of	—	—
Potassium and protein		
Sex characters in ~~male~~		
	Addison's disease ↳ low ~~bp~~ blood sugar, skin pigmentation, weight loss	Cushing's syndrome ↳ high blood sugar osteoporosis obesity.
- By promoting glucose utilization by body cells - By stimulating deposition of extra glucose (as glycogen) in liver and muscles	Diabetes mellitus ↳ hyperglycemia, glycuria with sugar, loss in weight	Hypoglycemia, Brains may enter a state of coma if sugar level becomes too low.
Stimulating the in the liver	—	—
Also influences development	Simple goitre (enlargement of thyroid) Cretinism (dwarfism & mental retardation in children) Myxoedema swelling of face & hands, sluggishness	Hyperthyroidism (Exophthalmic goitre) ↳ eyes are protruded, rapid heart beat, shortness of breath, increased metabolic rate

Gland	Hormone	Function of Hormone
PITUITARY		
(i) Posterior Lobe	Oxytocin	Causes contractions of uterine. It also stimulates milk
	Vasopressin (ADH)	Contracts blood vessels caus. Increases water reabsorption
(ii) Anterior Lobe	Growth hormone (Somatotropin)	Normal growth of the body
	TROPIC HORMONES	
	• Thyroid Stimulating Hormone. (TSH)	Stimulates thyroid to secret
	• Adreno-corticotropic Hormone (ACTH)	Stimulates secretion from
	• Gonadotropic Hormone	
	i) Follicle-Stimulating Hormone	Stimulates egg formation formation in males.
	ii) Luteinizing Hormone	Stimulates formation of ~~secretion of progestin~~ Stimulates secretion
	iii) Prolactin	Promotes lactation

	Undersecretion	Oversecretion
muscles during child birth. ejection.		
rise in blood pressure. from kidneys tubules.	Diabetes insipidus (frequent urination)	High blood Pressure
	Dwarfism	Gigantism (in childhood)
		Acromegaly (in an adult)
thyroxine		
adrenal cortex		
in females and sperm		
corpus luteum to produce progesterone, ovulation,		
of testosterone from testes		

Chapter 13

THE REPRODUCTIVE SYSTEM

Reproduction - Reproduction is the process of formation of new individuals by sexual or asexual means which can repeat the process in their own ~~term~~ turn

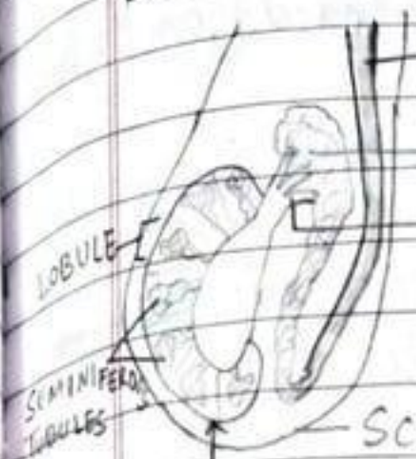
Asexual Reproduction	Sexual Reproduction
i) No gametes are formed	i) Male and female gametes are formed
ii) Only one parent is involved.	Usually two parents (male & female) are involved
iii) No ~~genetic~~ mixing of genes, therefore no (or little) variation in offspring.	Genetic mixing occurs, therefore increased variation in offspring
iv) It is rapid process in favourable conditions	It is slower process

Reproduction in Humans -

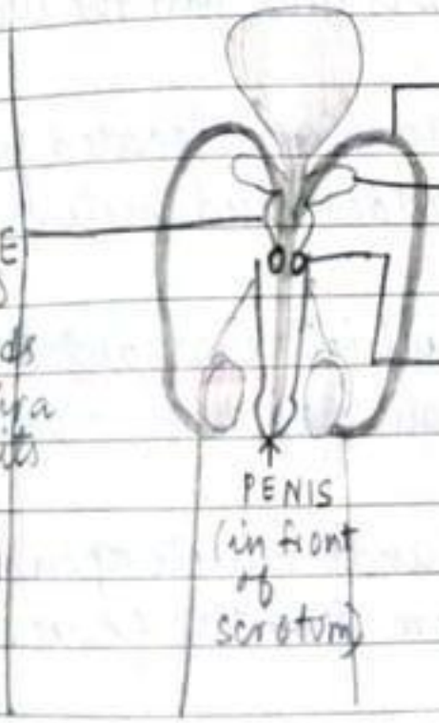
(1) PRIMARY REPRODUCTIVE PARTS - Ovary & Testes

(2) ~~SECONDI~~ ACCESSORY REPRODUCTIVE PARTS - Prostate gland, Cowper's gland etc.

Male Reproductive System



TESTIS -
- Seminiferous tubules (produce sperms)
- Leydig cells (packing tissue b/w coils of seminiferous tubules, produce testosterone)



ORGAN	LOCATION
① Testes	Inside scrotum suspended from the body
i) Seminiferous tubules	Inside lobules of testes
ii) Leydig cells	Packing tissue between coils of seminiferous tubules
iii) Epididymis	Tubular knot fitting like a cap on the upper pole of testes
② Sperm duct	Arises from testes to upward in the abdomen passing through inguinal canal. They loop over ureters and join the urethra.
③ Seminal Vesicles	Pair of lobulated glands located between posterior surface urinary bladder and rectum.
④ Prostate gland	Bilobed gland which surrounds the urethra close to its origin from the bladder.
⑤ Cowper's glands	Two small glands which open into the urethra just before it enters penis.
⑥ Penis	Cylindrical, vascular organ in front of the scrotum

FUNCTION

Testes produces sperms and secretes testosterone

To produce sperm by spermatogenesis

To produce testosterone

Stores the sperms for some days for maturation and to make them motile.

To carry sperm from testis to urethra

To produce a secretion (semen) which serves as a medium for transportation of sperms and ~~to~~ make sperms active

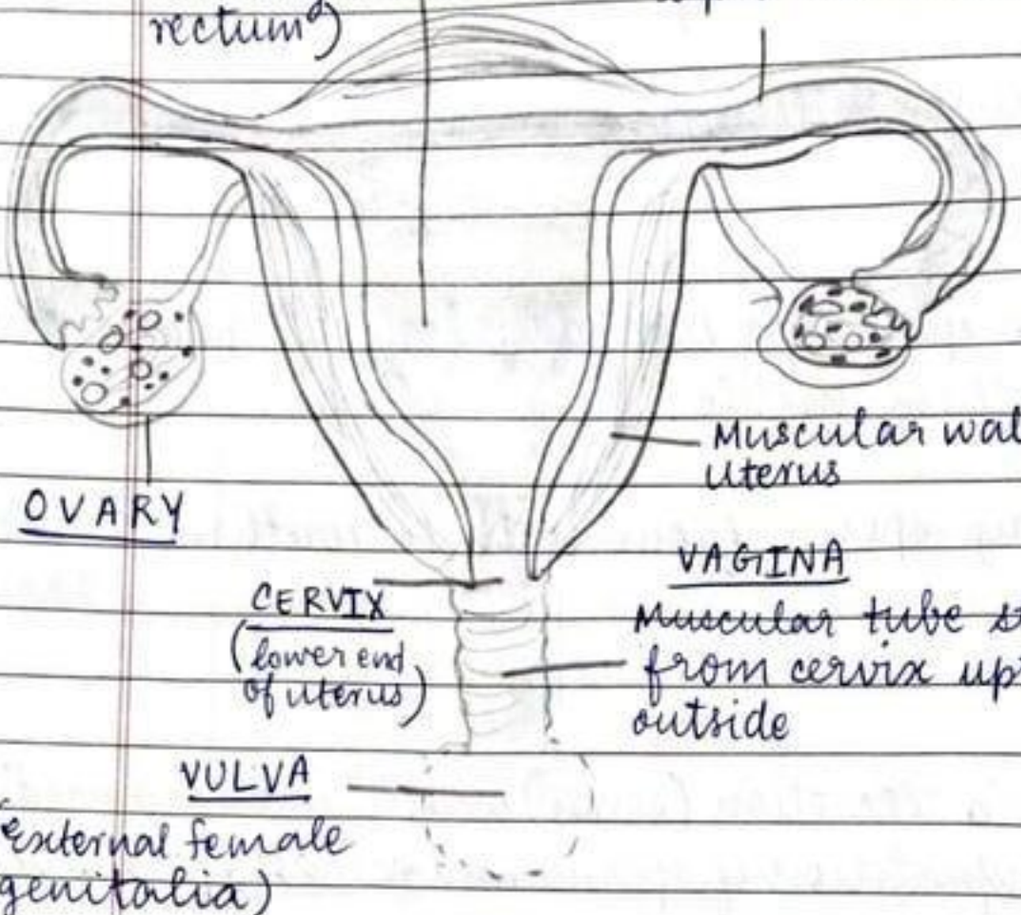
It pours an alkaline secretion into semen which neutralises acid in female's vagina.

To produce a secretion which serves as a lubricant of penis.

To transfer sperms into the female vagina

SHOTON LOPPO

Female Reproductive System



FUNCTIONS:

1. Ovaries – To produce ovum, oestrogen
   ↳ Corpus luteum: Secretes progesterone and oestrogen
2. Oviducts – • Transport ovum to uterus,
   • Site of fertilization of sperm & ovum
   • Zygote travels down via oviduct into the uterus.
3. Uterus – Protects and nourishes the growing embryo

SHOT ON OPPO

④ Vagina – • To receive sperms during copulation
• To allow passage of baby during childbirth.

⑤ Vulva – • To protect the urethra and vagina

MENSTRUAL CYCLE

Menarche – The onset of menstruation in a young female at about 13 years of age

Menopause – The permanent stoppage of menstruation in females at about 45 years of age.

Menstruation – The uterine bleeding caused due to shedding and rupturing of blood vessels of endometrium (thickened innermost lining of uterus)
It lasts ~~for of~~ about three to five days.

The period of a menstrual cycle is counted from the day of onset of flow to the next onset after 28 days.

SHOT ON OPPO

Menstrual Phase

~~Days~~ (3 to 5 days)

Blood is discharged due to shedding and rupturing ---- lining of uterus.

↓

Follicular Phase

Days 5 to 12 : follicle grows, inner lining of uterus thickens due to increased production of oestrogen.

↓

Ovulatory Phase

Days 13-14 : Ovulation occurs, egg travels down the oviduct

↓

Luteal Phase

Days 15-28 : Inner lining of uterus thickens further,

(The emptied follicle in ovary turns into ~~corpus~~ hormone-producing tissue called CORPUS LUTEUM which secretes progesterone & oestrogen)

DEFINITION OF CORPUS LUTEUM →

Definitions

1

(1) Fertilization – Fusion of the nucleus of male gamete and female gamete to form zygote is called fertilization. It occurs in the oviduct.

2

(2) Implantation – The process of fixing of the blastocyst to the wall of uterus (endometrium) is called implantation.

(3) Placenta – Placenta is the intimate connection established between the foetal membranes and the uterine wall, by which the foetus ~~embryo~~ gets food and oxygen and gives out carbondioxide and nitrogenous wastes into the mother's blood.

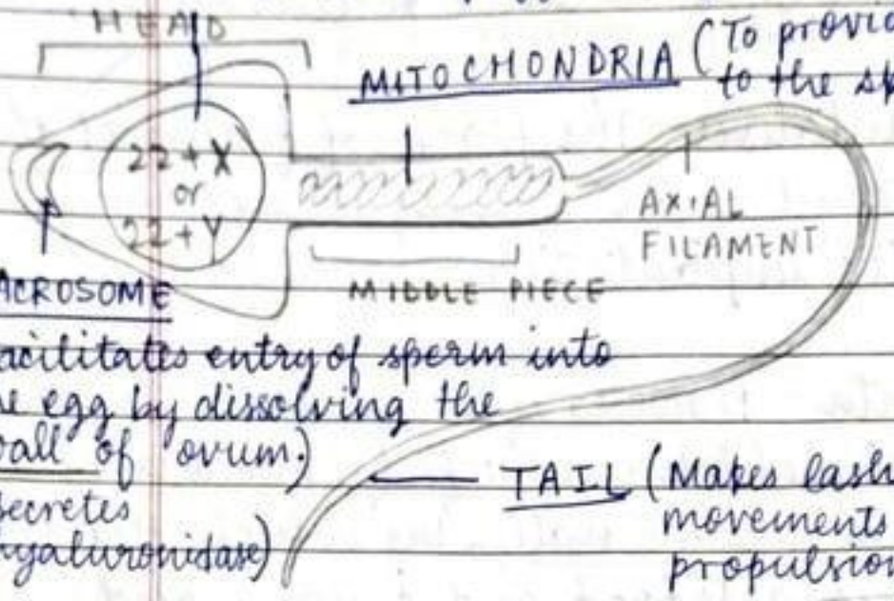
(4) Umbilical cord – The chord containing blood vessels, which connects the placenta with the foetus, is called umbilical cord

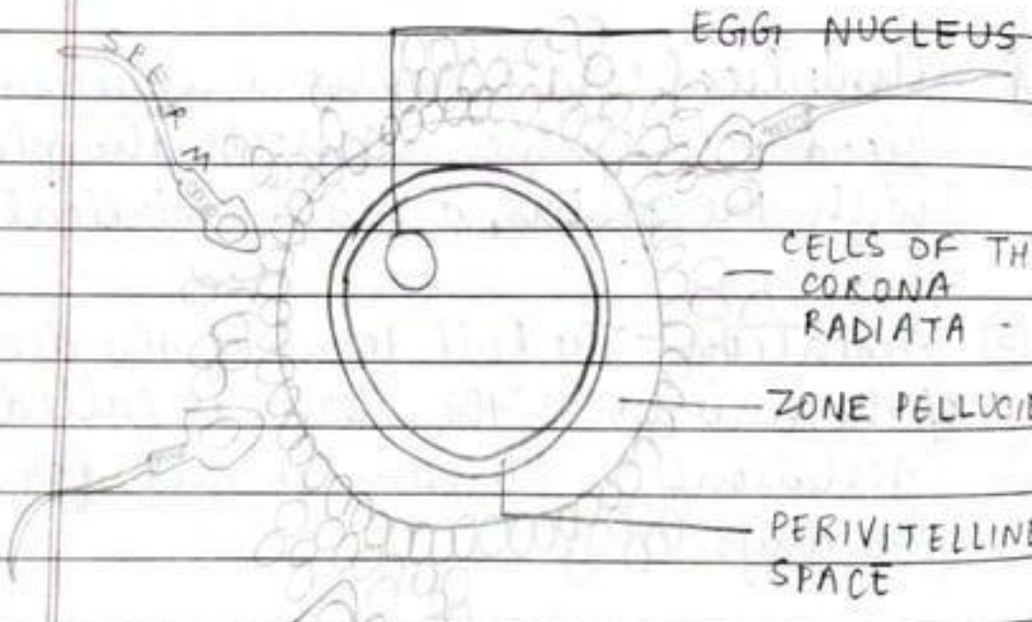
(5) Gestation – The full term of the development of the embryo in the uterus is called gestation. In humans, it lasts for about 280 days.

(6) Parturition – The act of expelling the full term foetus from the mother's uterus at the end of gestation is called parturition.

Sperm & Ovum : Structure and Functions

NUCLEUS
(Contains genetic material which combines with nucleus of egg during fertilization)





Role of Hormones in Reproduction

Puberty - The period of growth ~~in human~~ in which sexual maturity is attained is called puberty. For males, puberty occurs around 11 to 15 years of age whereas in females, puberty occurs around 10-14 years of age.

First sign of Puberty - Enlargement of breasts in femals and enlargement of testes in males.

What else - • Ovulation, secretion of oestrogen by ovary (in females).
• Secretion of Testosterone by testes (in males)

Role of

① Oestrogen -
- Growth of mammary glands
- Widening of hips
- Growth of hair in pubic region & armpits

② Testosterone -
- Deepening of voice
- Growth of penis and testes
- Growth of hair in pubic region, armpits, chest, chin and moustache.
- Regulates the formation of sperms.

③ Progesterone –
• Prepares the uterus for receiving the embryo

Pregnancy –

Embryo – the blastocyst after implantation in the uterine wall continues to grow and is called an embryo.

The embryo is about the size of a large pea and hardly resembles human being.

foetus – When embryo starts resembling human baby (8 weeks), it is called FOETUS

Amnion and amniotic fluid –

Amnion is a sac which develops around the embryo even before the formation of allantois. A fluid (amniotic fluid) fills the space between amnion and embryo.

Functions of amniotic fluid –

i) It protects the embryo from physical damage by jerks or mechanical shocks.

ii) Allows the foetus some restricted movement.

iii) Prevents sticking of the foetus to the amnion.

iv) Keeps an even pressure all around the embryo.

Placenta –

The intimate connection established between the foetal membranes and uterine wall, by which the foetus gets food and oxygen and gives out CO_2 and nitrogenous wastes into the mother's blood

Functions of Placenta –

i) Allows diffusion of glucose, oxygen, lipids, amino acids etc from mother to foetus

ii) Allows diffusion of CO_2, urea from foetus to mother.

iii) Produces the hormones oestrogens and progesterone.

NOTE – Placenta does not allow passage of germs from mother to foetus.

If there was a direct continuity of blood between the mother and foetus, the relatively high pressure of mother's

blood would seriously damage the soft delicate tissues of the foetus.

Twins

Fraternal Twins – Sometimes two eggs are ~~produ~~ released from ovaries at a time and both may get fertilised to produce two individuals.

Such twins produced from two eggs are called fraternal twins.

Identical Twins – Sometimes a single fertilized egg may split into two parts during its early stages ~~of cell division~~. Each of the two split parts behaves as an independent egg and produce one individual each.

Such twins produced from a single fertilized egg are called identical twins

Fraternal Twins are not alike because they come from different eggs and sperms.

Identical Twins look exactly alike because they come from same egg and sperm.

THE NERVOUS SYSTEM

Question / Answers

i. Short Answer Type

Q.1) Mention where in the human body are the following located and state their main functions:

a) Corpus callosum - located in the forbrain at the base of cerebrum, connecting the two cerebral hemispheres.
Function - To transfer information from one hemisphere to the other.

b) Central canal - located in the centre of spinal cord and runs the entire length of the spinal cord.
It is filled with cerebrospinal fluid which ~~acts as shock absorber and~~ protects the brain from external mechanical injuries and supplies nutrients to neurons.

Q.4) State the functions of the following:

a) Association neuron They interconnect sensory and motor neurons.

c) Medullary sheath – It provides insulation to neuron and helps in faster conduction of nerve impulses.

d) Cerebrospinal fluid – It ~~acts as a shock absorber~~ protects the brain from external mechanical injuries and supplies nutrients to neurons

Q.5) Rearrage the following in correct sequence pertaining to what is given within the brackets at end:

a) Effector – Sensory neuron – Receptor – Motor neuron – Stimulus – Central nervous system – Response (REFLEX ARC)

~~Receptor – Sensory neuron~~

Stimulus – Receptor – Sensory neuron – Central nervous system – Motor Neuron – Effector – Response

b) Repolarisation – Depolarisation – Resting (polarised) (CONDUCTION OF NERVE IMPULSE)

Resting (polarised) – Depolarisation – Repolarisation

c) Axon endings - Dendrites - Axon - Perikaryon - Dendron (NEURON STRUCTURE)
Dendrites - Dendron - Perikaryon - ~~Perikaryon~~ - Axon - Axon endings

d) Diencephalon, Cerebellum, Medulla oblongata, Pons, Cerebrum, Midbrain (SEQUENCE OF PARTS OF HUMAN BRAIN)
Cerebrum - Diencephalon - Midbrain - Cerebellum - Pons - Medulla oblongata

Q.6) Name the following :

a) 3 types of neurons -
Sensory, Moter, Association Neurons

b) 3 types of nerves -
Sensory, Motor, Mixed Nerves

c) 3 main parts of neuron -
Dendrites, Cyton, Axon

d) 2 major divisions of nervous system -
Central & Peripheral Nervous System

e) 3 layers of meninges -
Dura mater, Arachnoid, Pia Mater

f) 3 main parts of brain
cerebrum, cerebellum, medulla oblongata

g) 2 parts of the autonomic nervous system
Sympathetic and Parasympathetic

h) 2 types of reflexes
Inborn Reflexes and ~~for~~ Acquired Reflex

Q.7) Write the full form:

a) CNS Central Nervous System

b) ~~fn~~ CSF Cerebrospinal Fluid

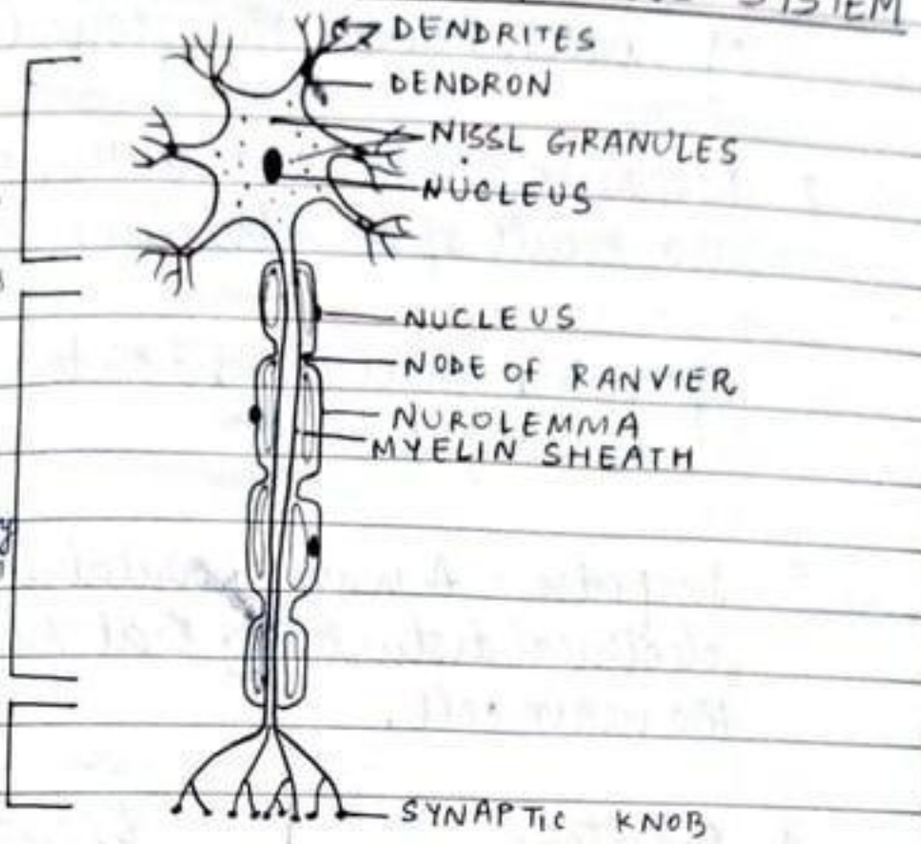
c) PNS Peripheral Nervous System

d) ANS Autonomic Nervous System

NOTES

The nervous system • keeps us informed of the outside world, • helps us to think, remember and to reason out, • controls voluntary activities and • regulates involuntary activities.

NEURON – UNIT OF NERVOUS SYSTEM



Conduction of Nerve Impulse through a neuron



(1) RESTING REGION (POLARISED)

The outer side of nerve fibre / fibre carries positive charge due to more Na^+ ions outside the axon membrane.

(2) DEPOLARISATION

On stimulation, the axon membrane at that spot becomes permeable and Na^+ ions move inwards. The point of depolarisation becomes a stimulus for next region of membrane, which in turn become depolarised

Definitions

1. Stimulus – An agent or a sudden change in the external or internal environment of an individual which induces a response in the organism or any of its body parts.

   eg – heat, hunger, pain, sound

2. Response – The change in the organism as a result of a stimulus is called response.

   eg – withdrawal of hand on touching a hot object.

3. Impulse – A wave of irritability, ie, electrical disturbance that sweeps over the nerve cell.

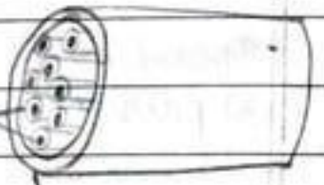
4.

Receptors	Effectors
• Receptors are specialised epithelial cells present in sense organs.	• Effectors are muscles or glands.
• On receiving a stimulus, receptors set up waves of impulses towards the central nervous system.	• On receiving the impulse from the central nervous system, effectors contract or secrete substances

5. Synapse – Synapse is the junction between the terminal branches of ~~an~~ axon of one neuron and the dendrites of another neuron, separated by a fine gap.

SENSORY NEURONS	MOTOR NEURONS	ASSOCIATION NEURONS
They carry the impulse from receptors (sense organs) to the central nervous system.	They carry the impulse from the central nervous system to the effector (muscle or gland)	They are located in the brain and spinal cord and they interconnect sensory and motor neurons.

6. Nerve – Nerve is a bundle of axons (nerve fibres) of separate neurons, ~~was~~ enclosed in a tubular sheath.

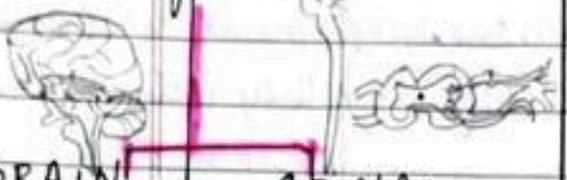


SENSORY NERVES	MOTOR NERVES	MIXED NERVES
They contain only sensory nerve-fibres.	They contain only motor nerve-fibres.	They contain both sensory and motor nerve-fibres.
They carry impulses from receptors (sense organs) to the central nervous system.	They carry impulses from the central nervous system to the effector (muscle or gland)	

7. Ganglion - It is an aggregate of cytons (cell bodies) of separate neurons from which axons may arise or enter into

NERVOUS SYSTEM

Central Nervous System



BRAIN

- Cerebrum - Site of memory, intelligence, consciousness and will power
- Cerebellum - Maintains balance and coordinates muscular activity
- Medulla oblongata - Controls involuntary actions of internal organs.

SPINAL CORD

- Controls reflex action
- Conducts sensory impulses from skin and muscles to brain
- Conducts motor impulses from brain to muscles of trunk & limbs

Peripheral Nervous System

SOMATIC NERVOUS SYSTEM

Conveys info to voluntary (skeletal) muscles

- Cranial nerves - 12 pairs
- Spinal nerves - 31 pairs *

AUTONOMIC NERVOUS SYSTEM

Control the involuntary action of internal organs

It contains a pair of chains of nerves and ganglia on either side of backbone

* every spinal nerve is a mixed nerve. Its distribution -
- 8 pairs - Cervical
- 12 pairs - Thoracic
- 5 pairs - Lumbar
- 5 pairs - Sacral
- 1 pair - Coccyx region

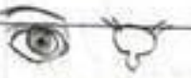


Sympathetic

its nerves arise from spinal cord between neck and waist

Function - This system prepares the body for action against abnormal conditions.

Parasympathetic



its nerves arise anteriorly above the neck and posteriorly below sacrum.

Function - It reestablishes normal conditions after violent act is over.

Ch- Pollution

Pollution is the contamination of air, water or land by substances which deteriorate the ~~natural~~ quality of ~~or~~ environment.

Wastes

Biodegradable
can be broken down by microorganisms into harmless and ~~not~~ non-toxic substances.

Non-biodegradable
cannot be broken down by micro-organism

Types	Sources
① Air	Vehicular exhaust, industries burning garbage, brick kiln
② Water	(liquid waste from domestic activities) Sewage, effluent discharged by industries, oil spills
③ Soil	Industrial waste (flyash, chemical residues etc), Biomedical wastes (needles, syringes, tablets etc), Pesticides, Chemical Fertilizers Domestic wastes

④ Radiations	X rays, Radioactive fallout from nuclear power plant
⑤ Noise	Train, loudspeakers, construction sites.

✡ Swachh Bharat Abhiyan — Launched on 2 October 2014 with the aim:

- to clean the roads and infrastructure of cities and towns
- to make India ODF by constructing toilets in rural India
- to achieve efficient solid and liquid waste management systems.

MCQ

① Ozone Day — 16th September
② DDT = Dichloro diphenyl trichloroethane
③ Particulate lead causes: Blood cell shortage
④ BS = Bharat Stage
(These norms mean that any motor vehicle running on road has to follow a specific emission norm to reduce pollution.)

MCQs

chapter - Pollution

1. Accumulation of a large amount of _____ waste leads to pollution.
Ans - non-biodegradable

2. Air pollution is of two types - gaseous and _____ pollution.
Ans - particulate

3. The ~~co~~ exhaust given out by automobiles contains three main gaseous pollutants :
Ans - CO_2, SO_2, CO

4. Harmful effect of smog in humans :
Ans - asthma and allergies

5. Brick kilns produce _____ which cause particulate pollution.
Ans - large quantities of ash, ~~broot~~ broken brick bits and smoke

6. _____ is the liquid waste from household activities.
Ans - Sewage

7. Sewage consists of _____, _____ and other household wastewater.
Ans- Kitchen waste, toilet

8. _____ is the liquid waste produced by factories.
Ans- Effluent

9. _____ pollution is localised whereas air and _____ pollutions can spread to long distances.
Ans: soil, Water

10. DDT, a pesticide stands for _____
Ans. dichloro-diphenyl-trichloroethane

11. Iodine ^{131}I increases the risk of _____
Ans. thyroid cancer

12. Gases responsible for acid rain are _____
Ans. CO_2, SO_2 (and NO_2)

13. Gases responsible for global warming are _____.
Ans. CO_2, CH_4

14. Gases responsible for ozone depletion are

Ans. Chlorofluorocarbons (CFCs)

15. September 16 is celebrated every year as International Ozone Day.

16. The BS emission standards are set up by:
Ans. Central Pollution Control Board
(Central Government)

Chapter- Population

1. Which day is celebrated as World Population day?
Ans. July 11

2. Name the first animal to be domesticated.
Ans. Dog

3. Population explosion refers to the rapid and massive rise in world population.

4. 2.1%, the highest annual growth rate ever recorded, was in the year _____
Ans. 1962

5. _____ means which can be continued with no ill-effect on the availability of required items.
Ans. Sustainable

6. Statistical study of human population with reference to size, density and distribution is termed as _____.
Ans. Demography

7. Population density is the number of individuals per _____.
Ans. square kilometre

8. _____ has highest population density with 407 persons / km^2.
Ans. Netherlands

9. Natality and Mortality are measured per _____ people of the population per year.
Ans. 1000

10. The inverted red triangle is a sign for _____.
Ans. Family Welfare

11. Round latex caps with coil springs, fitted on the mouth of cervix are called _____
Ans. Diaphragms

12. Devices which do not stop fertilization but prevent implantation of blastocyst are _____.
Ans. Intra-Uterine devices (eg. Lippie's loop & Copper T)

13. MTP stands for _____
Ans. Medical Termination of Pregnancy

14. Abortion is legally permitted till ___ months of pregnancy.
Ans. five

15. Hormonal Pills for contraception prevent

Ans. Release of the egg from the ovary

Ch- Structure of Chromosomes, Cell cycle and Cell Division

1. Each chromosome contains one long DNA molecule.

2. The complex of DNA and proteins is called chromatin.

3. Chromosomes were first studied in 1882 by Walther Flemming.

4. The shape of DNA molecule was first studied in 1953 by Rosalind Franklin

5. Each DNA strand is composed of repeating nucleotides

6. DNA strand wound around histone octamer is called nucleosome.

Date
Page

Absorption by Roots

1. Root hairs are extensions of cell of the root.
Ans. epidermal

2. The measure of tendency of a solution to take in water by osmosis is called:
Ans. Osmotic pressure

3. The recovery of from plasmolysis is called
Ans. deplasmolysis

4. The mineral ions enter the root hair cell by
Ans. Active transport

5. The pressure exerted by cell contents on cell wall is _____
Ans. Turgor pressure

6.

EYES AND EARS

ch- Sense Organs

1. Eye can be rotated with the help of six muscles.

2. Eyelashes prevent falling of larger particles into the eye while Eyebrows prevent rain drops (or sweat) from getting into the eyes.

3. Lysozyme is the antiseptic enzyme present in tears which kills the germs.

4. The cornea is a part of the sclera

5. The Choroid layer is richly supplied with blood vessels and contains melanin. It is one of the layers of eyeball. Iris is an extension of it.

6. Function of ciliary body (containing circular muscles) is: To alter the shape of lens.

7. The iris contains radial muscles to widen and circular muscles to constrict the pupil.

8.

YELLOW SPOT	BLIND SPOT
It is the area of brightest vision.	It is the area of no vision.
It has maximum number of sensory cells (cones)	It has no sensory cells, but is the point at which the nerve fibres bundle together and leave the eyeball (in the form of optic nerve)
It is located at the back of the eye almost at the centre in the horizontal axis of the eyeball. (In retina)	It is located lateral to the yellow spot on the ~~bot~~ nasal side. (In retina)

9. Define accomodation – The process of focusing the eye to see objects at different distances is called accomodation.
It is brought about by making the elastic lens flatter or thinner.

AQUEOUS HUMOUR	VITREOUS HUMOUR
Present between lens and cornea.	Present behind the lens, in the cavity of the eyeball.
Function: i) To keep the lens moist and protect it from physical shock.	Function: i) To maintain the shape of the eyeball
ii) To refract light.	ii) To protect the retina and its nerve endings

Give the function of -

1. Melanin in choroid layer -
It is dark black pigment which prevents light rays from reflecting and scattering inside the eye.

2. Ciliary body (present at junction of choroid and iris)
To alter the shape of the lens to see objects at different distances.

3. Suspensory ligament -
To hold the eye lens in position

Date
Page

4. Iris Regulates the amount of light entering the eye by adjusting the size of pupil.

5.

POPULATION

Reasons for sharp rise in population

① IN WORLD

- Tool making revolution
- Agricultural Revolution
- Scientific and Industrial Revolution

Population explosion refers to the rapid and massive rise in the world population that has occured over the last few hundred years.

Six main reasons for ~~te~~ sharp rise in population in the recent past -

i) Better healthcare facilities
ii) Fewer deaths due to vaccination programmes and more patients get cured.

iii) Fewer infant deaths due to better health care of newborn babies

iv) Large scale immunisation

v) Improved nutrition of growing children

vi) Food shortage minimised and green revolution. There are very few starvation deaths.

(2) IN INDIA

i) Traditional beliefs - Children are regarded as a gift of God and people of the lower strata make no efforts to avoid pregnancy.

ii) Desire for male child - It often leads to getting several children before getting one son.

iii) Economic reasons - Children are considered to be helping hands to increase family income.

iv) Illiteracy - Most of the rural population is ~~mostly~~ illiterate, superstitious and does not ~~#~~ know the functioning of human reproductive system.

v) Lack of recreation.

Problems faced due to population explosion -

i) Over-exploitation of natural resources such as forests, land, minerals and water. leading to their shortage

ii) Low per capita income

iii) Pollution of air, water and soil due to industrialization. It causes health problems.

iv) Unemployment and unequal distribution of wealth.

Terms related to Population

1. Demography
It is the statistical study of human population with reference to *
- size and density
- distribution
- other statistics

2. Population density
It is the number of individuals per square kilometre at any given time.
e.g., Netherlands has a population density of 407 persons / km^2

3. Birth rate - (Natality)
It is the number of live births per 1000 people of population per year.

4. Death rate - (Mortality)
It is the number of deaths per 1000 people of population per year.

5. Growth rate -
It is the difference between the birth rate and the death rate.

Methods of Population Control

1. Hormonal pills: e.g., Mala D
To prevent ovulation

2. Barrier methods:
↳ Males - Condom
- To prevent sperms from being deposited in the vagina

↳ ~~Diaphragms~~ Females
1 - Diaphragms
- Round latex caps with coil spring
- Prevent entry of ~~germ~~ sperms into the uterus.

2 – Spermicidal agents
- Chemicals placed in the vagina near cervix
- Kill sperms if they are there

3. Intra-Uterine Devices : e.g., Lippe's Loop and Copper-T
- Fitted inside uterus
- Do not ~~prevent~~ stop fertilization
- Prevent implantation of blastocyst

4. Surgical Methods :

↳ Males – Vasectomy
- Vas deferens from each testes is ligated (tied with a nylon thread) and a small piece between the two ligatures is removed.

↳ Females – Tubectomy
- Abdomen is opened and the fallopian tubes are ligated (tied with a nylon thread) and a small part ~~of~~ b/w the ligatures is removed.

5. MTP (Medical Termination of Pregnancy)

- If the couple does not want to have the child, or
- If there is definite evidence of any serious genetic disease in the embryo.

Abortion is ~~only~~ legally permitted only within five months of pregnancy.

'Gene' term was coined by Wilhelm Johanssen in 1909

Haemoglobin contains iron and protein

The average life of RBC is 120 days.

Abnormal increase in the number of RBCs - Polycythaemia

Abnormal decrease in the number of RBCs. - Erythropenia

Average life of WBCs - 2 weeks
Average life of platelets - 3-5 days

Antigens are specific proteins present in on surface of RBCs.
Antibodies are complementary proteins present in blood plasma

Vasodilation and vasoconstriction occurs in capillaries and arteries but NOT in veins

Examples of diuretics are tea, coffee alcohol, liquid diets. (Substances which increase formation of urine

N.01.23

RANDOM KNOWLEDGE BITES

(from Previous Biology Question Papers ICSE)

Main functions –

1. Cerebrospinal fluid – It protects the brain from mechanical injuries
2. Eustachian tube – It equalizes air pressure on both sides of the tympanum to allow it to vibrate freely.
3. Suspensory ligament of eye – To hold the eye lens in position.
4. Sperm duct – To transport sperms from testis (epididymis) to the urethra
5. Lenticels – Diffusion of oxygen and carbon dioxide. Removal of water vapour during transpiration
6. Lymphocytes – They produce antibodies

7. Leydig cells – The secrete testosterone

8. Corpus luteum – It secretes progesteron and oestrogen and prepares the uterus for implantation of zygote.

Exact location –

10. Centromere – Region joining sister chromatids, region of ~~of~~ chromosome to which spindle fibre is attached during cell division.

11. Chordae tendinae – ~~Chor~~ Tendinous chords arising from papillary muscles of ventricles, which hold the flaps of bicuspid and tricuspid valves in position.

12. Proximal convoluted Tubule – In the cortex region of kidney. Part of nephron between ~~Bowr~~ Malpighian capsule and Henle's loop.

13. Ciliary body – Extension of choroid of the eye.

Exact location

1. Epididymis – Tabular knot fitting like a cap on the upper pole of ~~testes~~ testis
2. Lacrimal gland – Upper sideward portion of the orbit of eye / Upper outer corner of the eye
3. Malleus – In the middle ear between eardrum and incus.
4. Hydathodes – Pores on the margins of leaves
5. Pulmonary semilunar valves – In the right ventricle at the base of pulmonary artery.
6. Amnion – Sac surrounding/enclosing the embryo in the uterus.
7. Yellow spot – Centre of horizontal axis of eye ball on the retina
8. Organ of corti – On the basilar membrane of median canal in cochlea, present in inner ear.
9. Loop of Henle – Part of nephron located in renal medulla of kidneys.
   ↑ between PCT and DCT

Define / Name the following:

1. Homologous chromosomes – A pair of corresponding chromosomes of the same shape and size, ~~deriv~~ derived one from each parent.

2. The layer of eyeball which provides nourishment to the eye – Choroid

3. The layer of eyeball which forms the transparent cornea – Sclera

4. Cuticle – The transparent waxy layer on the epidermis of leaves which reduces transpiration is called cuticle

5. The organ which produces urea – liver

6. In the S-phase of cell cycle, what is synthesized more? DNA

7. Crossing over – The exchange of chromatid segments between the maternal and paternal chromatids of a pair of homologous chromosomes, during meiosis is called crossing over
   ↑ or genetic recombination

8. Mineral ion needed for the formation of blood clot – Calcium

9. Complete the paragraph –

(i) The amount of urine output is under the regulation of a hormone called _____ secreted by the _____ lobe of the pituitary gland. If this hormone secretion is reduced, there is an increased production of ~~the~~ urine. This disorder is called _____. Sometimes excess glucose is passed with urine due to hyposecretion of another hormone called _____ leading to a disease called _____.

ANSWER : Antidiuretic hormone
Posterior
Diabetes insipidus
Insulin
Diabetes mellitus

(ii) To test a leaf for starch, the leaf is boiled in water to _____. It is then boiled in methylated spirit to _____. The leaf is dipped in warm water to soften it. It is placed in a petri dish, and _____ solution is added. The region of leaf which

contains starch, turns _____.
The region which does not contain starch, turns _____.

ANSWER: Kill the cells
remove chlorophyll
Iodine
blue-black
yellowish brown.

classmate
Date
Page

MITOSIS

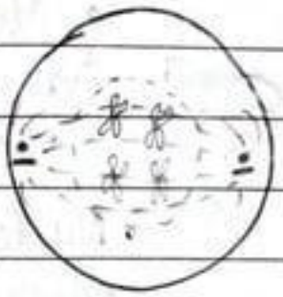
Cell division in which two identical daughter cells are formed from the division of one parent cell.

Purpose - growth, repair, replacement

2 phases - Karyokinesis
- Pro
- Meta
- Anaphase
- Telophase

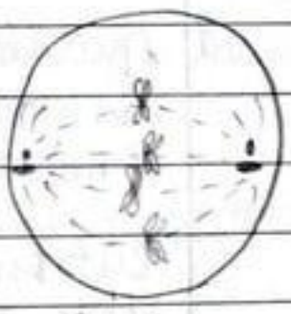
Cytokinesis → furrowing of cytoplasm OR cell plate formation

① Prophase



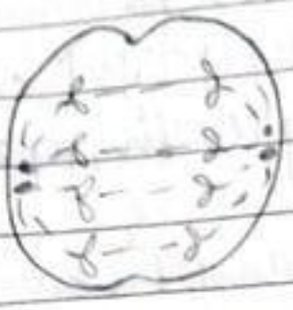
- Chromosome is condensed
- Nuclear membrane and nucleolus disappear
- Spindle fibres are formed
- Centrosome splits and duplicates and occupies poles of the cell.

② Metaphase



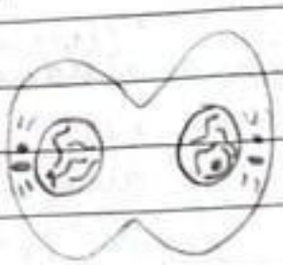
- Chromosomes align in the equatorial (metaphasic) plane
- Spindle fibre attaches to the centromere of each chromosome

③ Anaphase



- Centromere splits
- Sister chromatids are pulled apart towards opposite poles
- Spindle fibres disappears
- Furrow starts appearing (animal cell)

④



- Chromatids decondens to from chromatin fibre
- Nucleolus & nuclear membrane reappear
- Spindle fibres disappear

Mitosis in animal cells	Mitosis in plant cells
• Asters are formed.	Asters are not formed
• Spindle fibre is formed by asters and centrioles.	Spindle fibres are formed with the help of microtubules
• Cytokinesis by furrowing of cytoplasm.	Cytokinesis by cell plate formation

Interphase (Before and After Mitosis)

① G_1 (First Growth Phase)

- Volume of cytoplasm increases
- Mitochondria and chloroplasts divide
- RNA and proteins are synthesized
- After G_1 phase, cells may either enter a resting phase or enter S-phase

② S (Synthesis Phase)

- More DNA is synthesized each DNA molecule duplicates by semi-conservative replication of DNA. (i.e, DNA double helix opens at one end and new strands begin to form)
- Chromosome duplicates but remains attached

③ G_2 (Second Growth Phase)

- RNA and protein synthesis continues

This sums up the cell cycle –
The series of events which take place in a cell leading to the ~~duplication of DNA~~ duplication of its DNA and subsequent cell division to produce two daughter cells.

Genetics

The study of heredity i.e, transmission of body features from parents to offspring and the laws relating to such transmission.

Genetic counselling Newly married couples are advised to consult a specialist regarding the possibilities of any ~~des~~ undesirable trait which the children might inherit.

Genetic engineering - Genetic constitution of an organism is altered by introducing new genes into its chromosomes.

Heredity - Transmission of ~~body features from pro~~ genetically based characteristics from parents to offspring

Trait - Alternative form of character

Allele - Alternative forms of a gene occupying the same position on homologous chromosomes and affecting the same characters.

Sex~~ed~~ Linked Inheritance

Sex linked inheritance is the appearance

a trait which is due to the presence of an allele found exclusively on the X or on the Y chromosome.

Excretory System

- Lungs → remove CO_2
- Liver → produces urea, ammonia
  bilirubin (by breakdown of haemoglobin)
  breaks down alcohol, bad cholestrol, drugs.
- Skin → remove sweat (water + NaCl)

(Accessory excretory organs)

Primary
- Kidneys → excrete urea, excess salts

Kidneys

located on either side of backbone in the abdominal cavity protected by the last two ribs.

Production of Urine

① Ultrafiltration Filtrate (Blood plasma excluding proteins & blood cells) enters Bowman's capsule due to high pressure in glomerulus.

② Selective Reabsorption In PCT, most water and glucose and Na^+, Cl^- ions are reabsorbed. In Henle's loop water and Na^+ reabsorbed. In DCT, water and Cl^- ...

③ Tubular secretion walls of DCT secrete

K^+ ions and chemicals like ~~like~~ penicillin into the forming urine. ~~in t~~

↑

filtrate left after selective reabsorption and tubular secretion:

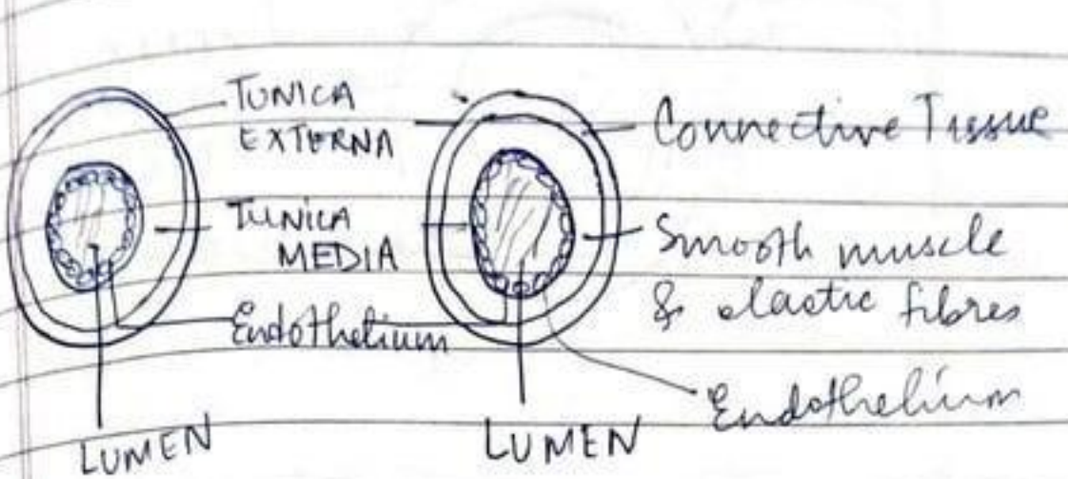
- yellow due to urochrome
- Slightly acidic
- 95% water
- 5% dissolved solid wastes:
  Urea, Creatinine, NaCl, KCl, Ammonia, Uric acid.

Stimulus → receptor → Sensory Neuron → CNS → Motor ~~Resp~~ Neuron → Effector → Response

cerebrum → Diencephalon → Midbrain
cerebellum → Pons → Medulla oblangata

circulatory

Pus - composed of dead WBCs together with tissue cells destroyed by bacteria.



Cells absorbs oxygen and other substances from tissue fluid.

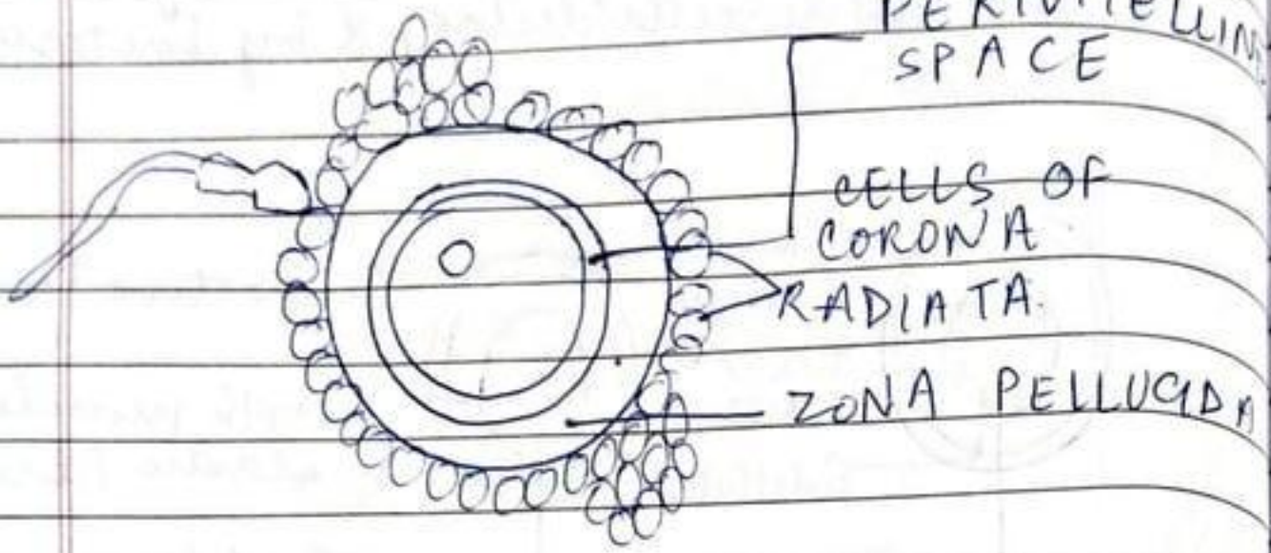
Lymph Nodes —

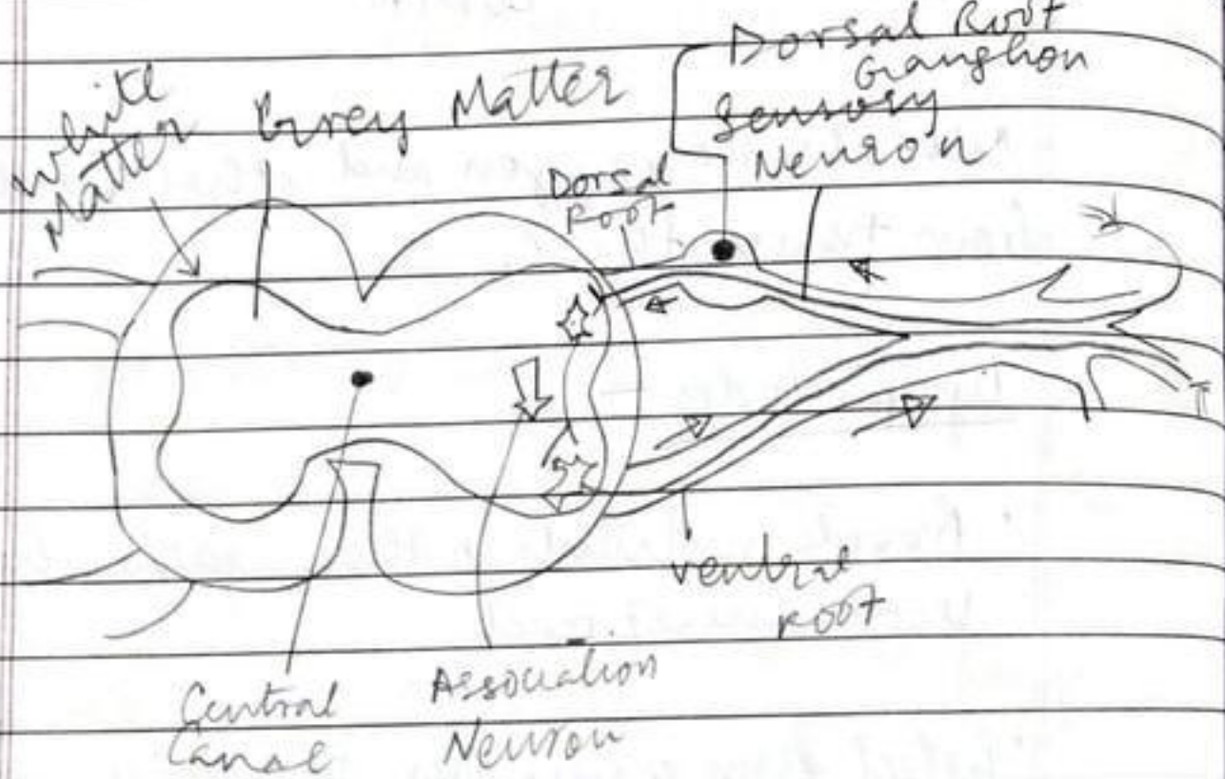
- Provide nutrients to those parts where blood cannot reach
- Protect from germs by producing antibodies

Spleen

① Produces lymphocytes
② Destroys worn out RBCs
③ Produces RBCs in embryo.

β cells of Islets of Langerhans → Insulin
α cells - Glycogen Glucagon





classmate
Date
Page

Define

① Reflex action – It is an automatic quick ~~respons~~ involuntary action in the body brought about by a stimulus.

②

Natural Reflex.	~~Unborn~~ Conditioned Reflex
They are reflexes requiring no previous experience.	They are reflexes developed by experience or learning
Similar in all human beings	It differs in different individuals subject to learning experience.

classmate
Date
Page

Long Question-Answers

① Application of Mendel's laws

② Use of Turgidity in Plants

i) To provide rigidity to soft tissues such as leaves

ii) To help the seeds push through the hard ground

iii) Turgor pressure in root hair cells builds up root pressure

iv) Turgidity helps in the opening and closing of stomata

v)

Inflammation - It is a reaction of the tissue to injury and localised infection. features of inflammation are - increased local heat, redness, swelling, pain.

Monocytes and neutrophils migrate through walls of blood vessels to fight disease causing germs.

Rough

Be

PRACTICAL

classmate
Date
Page

① Demonstration of diffusion (VISUAL)

Procedure -

i) Fill the beaker three-fourth with water
ii) Place $KMnO_4$ crystals in the beaker
iii) Leave the beaker undisturbed for sometime.

Observation -

The molecules of $KMnO_4$ start dissolving slowly in the water of beaker. They move from their high conc. to low concentration.

The water in the beaker becomes colourful.

Conclusion -

The molecules of $KMnO_4$ crystals move by the process of diffusion from their high conc. to low conc. Water in the beaker becomes uniformly colourful after sometime.

Diffusion is the free movement of molecules of a substance (solute or solvent, gas or liquid) from the region of their higher concentration to the region of their lower concentration when the two are in direct contact.

② Demonstration of diffusion (PROPERTY-WISE)

Procedure-

i) Drop a sugar cube in a beaker filled three-fourth with water.

ii) Taste a teaspoon full of water from the surface soon after starting the experiment.

iii) Taste the water repeatedly 4 to 5 times after an interval of 1 minute.

Observation-

Water is found to be sweeter with every tasting.

Conclusion-

The sugar molecules of the cube get gradually diffused into the medium, water, thereby making it sweet.

classmate
Date
Page

③ Demonstration of osmosis with the help of thistle funnel

Procedure -

i) Take a wide mouthed thistle funnel with a narrow stem.

ii) Cover the mouth of thistle funnel with cellophane paper and pour concentrated sugar solution in it.

iii) Invert this thistle funnel in beaker 3/4 filled with water.

iv) Mark the level of sugar solution on the stem of funnel as initial level.

Observation -

i) After a few hours, the level of liquid in the stem of thistle funnel rises up.

ii) If the water in the beaker is tasted it is not found to be sweet.

Conclusion -

Cellophane paper allows only water molecules to pass but not the sugar molecules. Thus cellophane paper acts as a semi permeable membrane.

Osmosis is the movement of only solvent molecules from the region of their higher concentration to the region of lower conc. across a semipermeable membrane.

(5) Demonstration of Root Pressure in Plants

Procedure-
Cut the stem of the plant and fix a glass tube with the help of a rubber connector.

Observation-
Cell sap starts oozing out from the cut end of the stem and it rises up in the glass tube, pushing the mercury of manometer.

Conclusion-
The pressure of absorbed and rising cell sap can be measured by a manometer. This pressure is called the Root Pressure.

Root Pressure is a pressure under which water passes from living cortical cells of root into xylem. It is built up due to cell to cell endosmosis and helps in ascent of sap.

⑥ Demonstration of Transpiration

Procedure -

Cover the experimental plant with a polythene bag and tie its mouth at the base of the stem.

Leave the plant in sunlight for an hour.

Control -

A polythene bag tied ~~to~~ at the base of a dead and dry stem is exposed to sunlight.

Observation -

In the experimental setup, water droplets appear on the inner wall of the polythene bag.

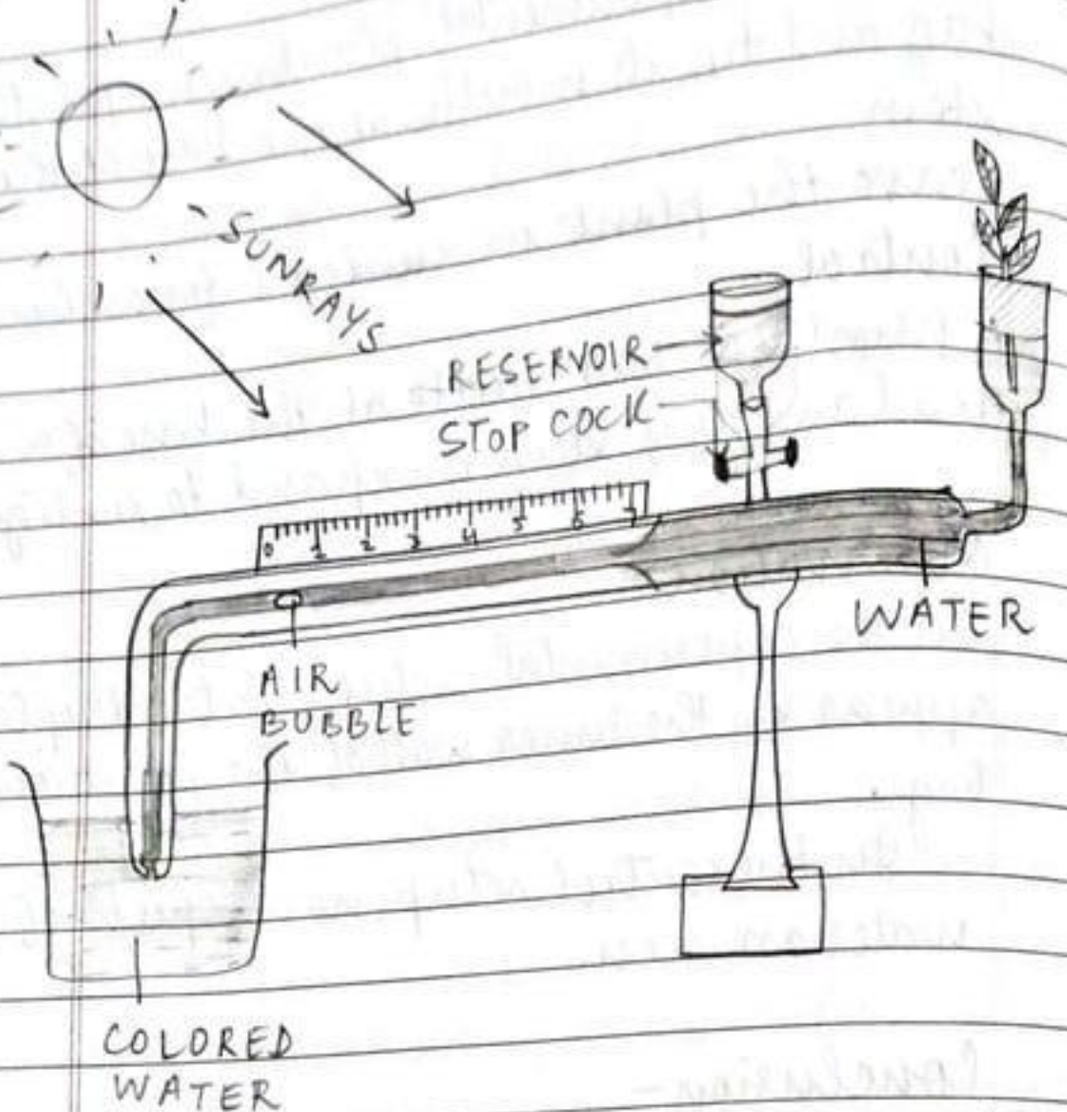
In the control setup, no droplets of water are seen.

Conclusion -

The aerial parts of a living and green plant give out water as water vapour in the daytime.

Transpiration is the process of loss of water in the form of water vapour from the leaves and other aerial parts of ~~the~~ a ~~plant~~ living and green plant.

⑦ To demonstrate rate of transpiration by using Ganong's potometer.

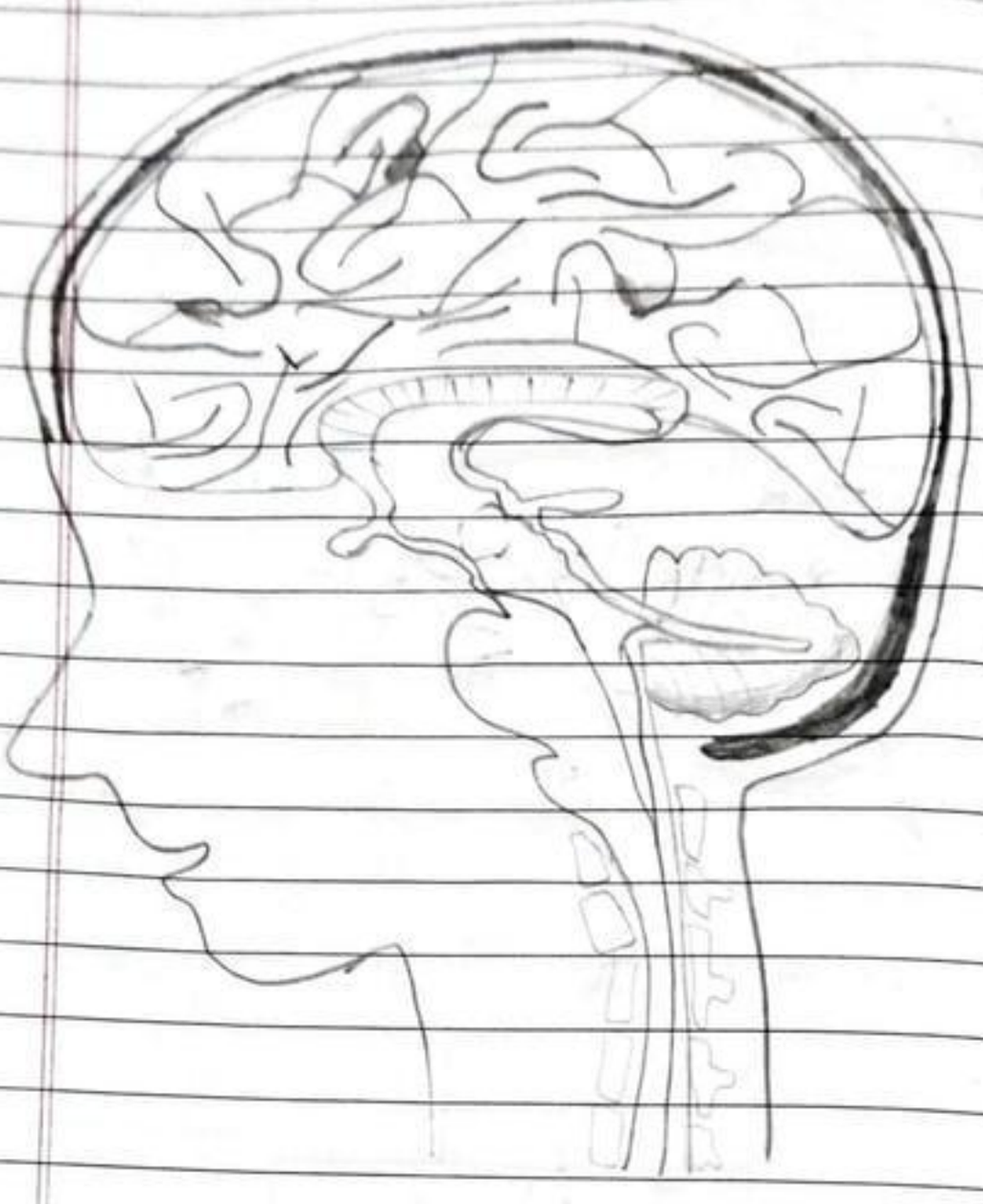


Precautions during the use of Potometer

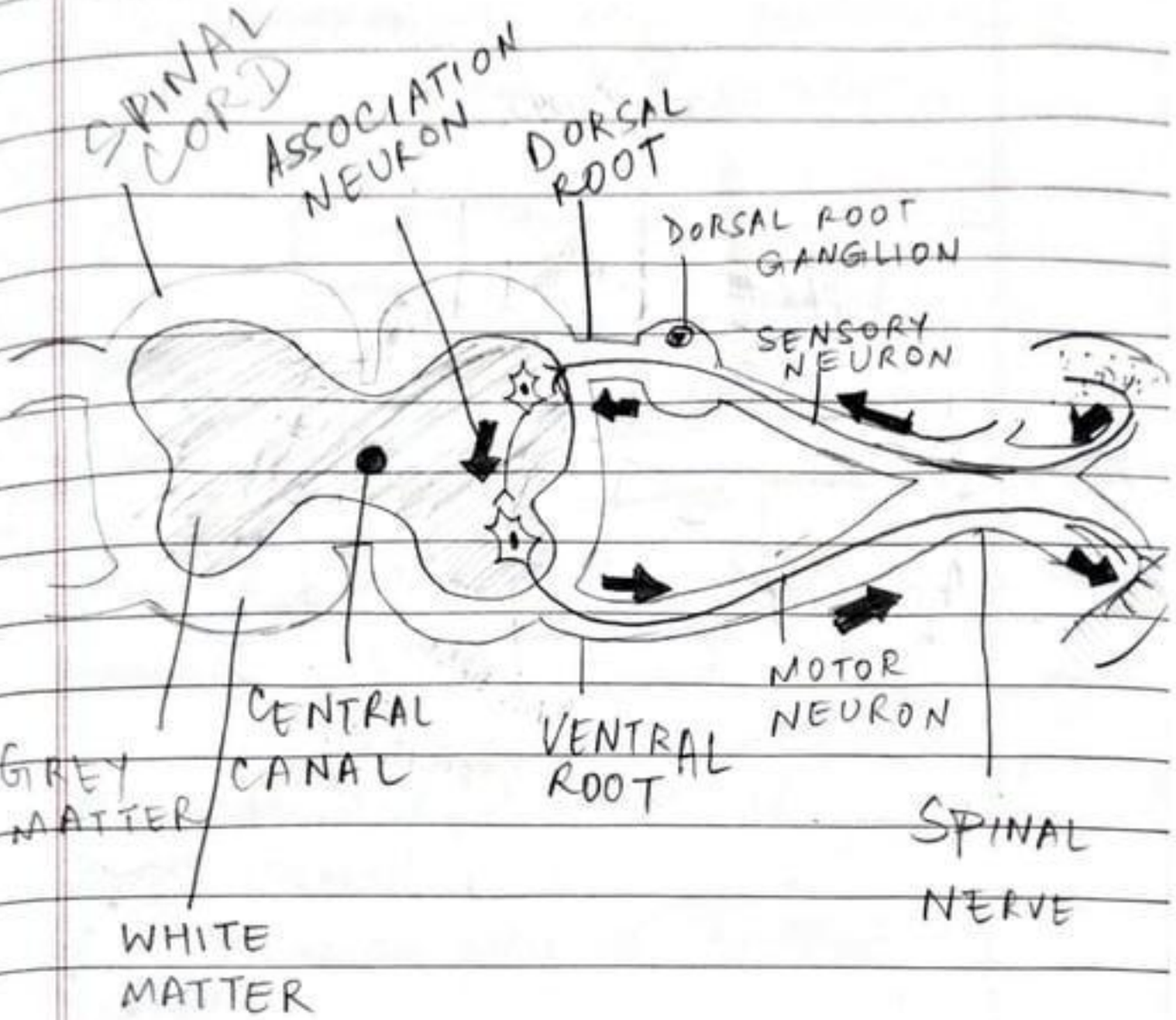
i) The potometer should be made completely water-tight
ii) The twig should be cut obliquely (to allow larger surface for water intake) and under water (to avoid suction of an air bubble into the twig which will stop the absorption of water into the xylem.)

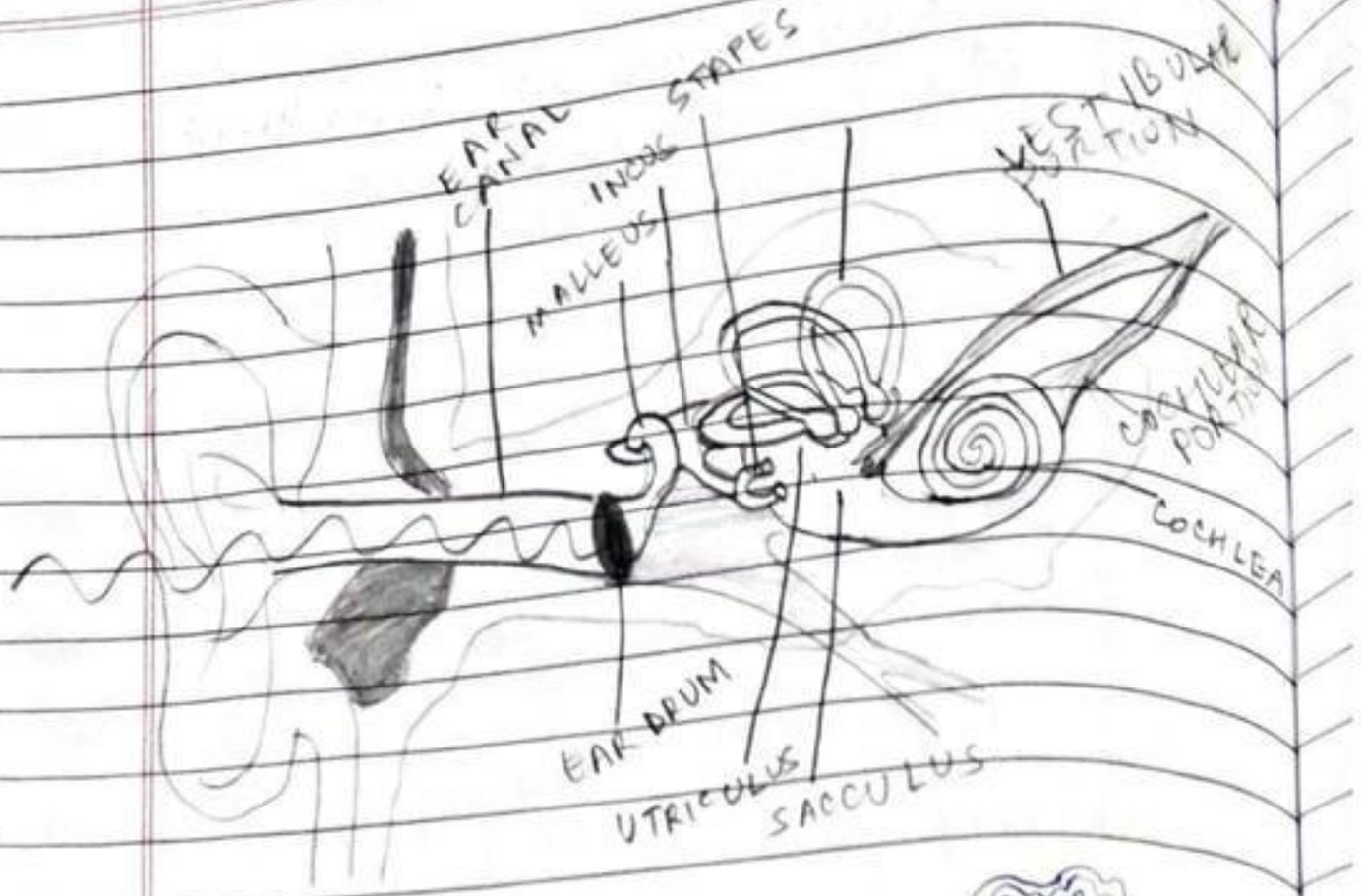


classmate
Date
Page

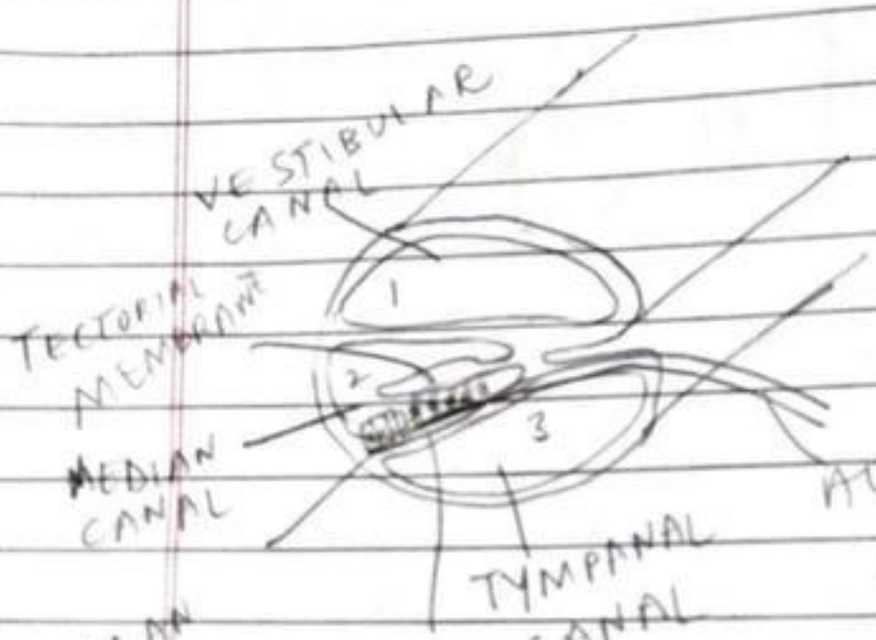


classmate
Date
Page




EAR CANAL
INCUS
STAPES
MALLEUS
VESTIBULAR PORTION
COCHLEA
EAR DRUM
UTRICULUS
SACCULUS




VESTIBULAR CANAL
TECTORIAL MEMBRANE
1
2
3
MEDIAN CANAL
TYMPANAL CANAL
AUDITORY NERVE FIBRE
ORGAN OF CORTI
BASILAR MEMBRANE